

Blast cell segmentation and leukemia classification using hybrid Deep Kronecker WideResNet using blood smear images

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ABSTRACT

Acute Lymphoblastic Leukemia (ALL) is a dangerous form of leukemia, which disturbs the bone marrow and irregular White Blood Cells (WBC) for the persons of all age groups. Early diagnosis of leukemia is necessary to provide appropriate care and cure the patients. Still, the detection of ALL from Peripheral Blood Smear (PBS) images is complicated due to the size and shape of the cells. This variability creates more difficulty for the segmentation and classification of Leukemia. To solve these issues, the Deep Kronecker Wide Residual Network (DKWRN) is developed for classifying Leukemia. The blood smear images are preprocessed using adaptive Gaussian filtering; hence, the noise from the image is diminished. After that, RefineNet effectively segments the Blast cell from the image. Then, augmentation such as flipping, resizing, and rotation enhances the dimension of the image. Following augmentation, essential features like Binary Pattern of Phase Congruency (BPPC) with Discrete Cosine Transform (DCT), and statistical features are extracted. Finally, the DKWRN classifies the Leukemia into early Pre-B, Pre-B, Pro-B, and Hematogones. Here, the proposed DKWRN is the integration of WideResNet (WRN) and Deep Kronecker Network (DKN). Furthermore, the proposed model offered optimal accuracy, True Negative Rate (TNR), recall, precision, and F1-score of 92.12 %, 91.40 %, 90.36 %, 91.56 %, and 90.96 %, respectively. Furthermore, the proposed DKWRN model demonstrated notable enhancements in classification accuracy when compared to existing techniques. Specifically, its performance exceeded that of Bayesian-based optimized convolutional neural network (CNN) for ALL detection (BO-ALLCNN) by 5.91 %, Support vector machine (SVM) by 4.63 %, ResNet-18 combined with SVM by 4.04 %, Residual Convolutional Neural Network (ResNet-152) by 2.27 %, DKN by 1.51 %, and WRN by 0.55 %, respectively.

1. Introduction

Medical diagnosis is considered a significant way to detect disease. Moreover, it detects diseases based on a patient's symptoms, diagnostic tests, and medical history (Das and Dutta, 2020). Microscopic analysis is considered the most effective method for testing blood cells. The utilization of microscopic images ensures the detection of all disorders. Moreover, blood transmits nutrients, energy, and oxygen to body cells while eliminating waste and Carbon-Dioxide (CO₂). In blood fluid, WBCs are termed as immunity or leukocyte cells, which are considered a

substantial part of the immune system. It is generated in the bone marrow and circulates throughout the body (Abrol et al., 2023). Hence, the prime step is to detect and diagnose the condition of blood smears contained in the WBCs, platelets, and Red Blood Cells (RBCs). Moreover, blast cells exist in the peripheral circulation among the regular blood cells (Le et al., 2015; Das and Dutta, 2020). The blast cells are termed as small WBCs present in the bone marrow and peripheral blood. It is a precursor cell, which is transformed into a mature WBC, such as a lymphocyte, monocyte, or neutrophil. In addition, the blast cells are categorized into three types depending on maturity and shape. The

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blood cells are grouped as pro myeloblasts, myeloblasts, and blast cells. The blast cells are much lighter than mature blood cells. The presence of blast cells in bone marrow is symptomatic of leukemia or other hematological illnesses.

Leukemia is a kind of deadly blood disorder (Vogado et al., 2021) that affects the immune system of the human body by generating cancerous WBC. It is more common in children and teenagers (Raina et al., 2024). Leukemia is produced by the abnormal propagation of leukocytes or WBCs in the bone marrow that do not respond to the cell-growth inhibitors (Le et al., 2015). Furthermore, it results the hematopoiesis suppression, anemia, thrombocytopenia, and neutropenia. Immature WBCs are typically found in the gonads, meninges, thymus, liver, lymph nodes, and spleen. Excess lymphoid blasts or myeloid cells in the marrow flow into the peripheral bloodstream and are the cause of leukemia (Das and Dutta, 2020). Leukemia begins in the bone marrow and rapidly spreads to other parts. Bone marrow is located inside the bone cavity and creates RBCs, WBCs, and platelets. Furthermore, blood cells are created from stem cells that are matured into myeloid or lymphoid cells before being distributed into the circulation (Bigorra et al., 2017). Leukemia develops when the lymphoid or myeloid cells begin to increase rapidly and uncontrolled. Lymphoblastic leukemia and Myeloid leukemia are the two kinds of leukemia, which are distinguished by the occurrence of myeloid or malignant lymphoid stem cells (Acevedo et al., 2019). Moreover, Myeloid and Lymphocytic leukemia are categorized into two subtypes, such as acute and chronic (Boldú et al., 2019; Raina et al., 2024). Leukemia patients are screened using a microscope, which examines the blood smears or microscopic slides. Domain specialists estimate the microscopic slides for making disease diagnoses. Still, this method is time-consuming and complex (Atteia et al., 2022).

Recently, there has been a lot of interest in Deep Learning (DL), which has been employed in a variety of healthcare applications (Khan et al., 2022; Samee et al., 2022). Convolutional Neural Network (CNN) is a popularly used DL model in Leukemia disease diagnosis. Moreover, CNNs achieve numerous disciplines for computerised medical applications. CNN-based Computer-Aided Diagnostic (CAD) has been recognized for its capability of identifying blood-related disorders (Atteia et al., 2022). Hence, CNN is a popular model in DL, particularly in image-related medical datasets, owing to its excellent adaptability, generalization, and self-learning properties. Moreover, CNN eliminates the requirement of handcrafted feature extraction; therefore, its self-learning capability carries out the intended task without any support (Mishra et al., 2019). A lot of DL-based algorithms have been demonstrated owing to their effective use in a variety of processes such as automated segmentation, detection, and classification of Leukemia (Perveen et al., 2024). DL models (Bhosale et al., 2022, 2024) outperform Machine Learning (ML) approaches in terms of accuracy and reliability on such tasks. These models automatically extract the features of 2-dimensional/3-dimensional(2D/3D) medical images during the segmentation, detection, and classification processes. Previously, various pre-trained DL models are utilized to assess more difficult and decisive illnesses, such as Acute Lymphoblastic Leukemia (ALL) (Saeed et al., 2024). Most of the flaws of manual analysis and medical imaging have been addressed by CNN, which offered a greater ability to distinguish the normal and blast cells (Abunadi and Senan, 2022). In addition, the hybrid network is a better choice for the diagnosis and classification of blast cells (Abunadi and Senan, 2022).

In this paper, the DKWRN is devised to classify Leukemia. Here, the multi-cell blood smear images gathered from the given dataset are applied to preprocessing, where the adaptive Gaussian filter is employed. Following that, the RefineNet segments the blast cells. Afterwards, the augmentation methods such as flipping, resizing, and rotation are performed. In feature extraction, the Binary Patterns of Phase Congruence (BPPC) with Discrete Cosine Transform (DCT) features, and statistical features are extracted. At last, the leukemia is classified into early Pre-B, Pre-B, Pro-B, and Hematogones by the

utilization of DKWRN.

The key contribution of this paper is explained below:

- **DKWRN-based Leukemia classification using multi-cell blood smear images:** In this work, Leukemia is classified using the DKWRN. Here, the DKWRN is developed by the merging of the WRN with the DKN.

The subsequent structure of the manuscript spans five key sections, beginning with Section 2, which explores the foundational rationale and surveys prevailing techniques used in leukemia classification. In Section 3, the DKWRN-based leukemia classification is illustrated. Section 4 shows the experimental result of DKWRN-based Leukemia classification. Section 5 describes the conclusion.

2. Motivation

Leukemia is created by the extensive propagation of blast cells. Hence, the segmentation of these cells is significant to validate the presence leukemia subtype. Because different classes and sub-classes of leukemia need dissimilar treatments. Therefore, the classification of leukemia is needed to provide the proper treatment at an earlier stage.

2.1. Literature survey

Atteia, G., et al. devised the Bayesian-based optimized convolutional neural network (CNN) for ALL detection (BO-ALLCNN). The model achieved high classification performance without relying on advanced image processing techniques such as augmentation and segmentation; however, its applicability was limited, as it could not be directly transferred to other medical imaging datasets (Atteia et al., 2022). Baig, R., et al. developed the hybridized CNN model for the detection of Malignant Leukemia. This model was applicable to detect leukemia cells with the minimal risk of human error and diminish the workload of diagnosticians. Nevertheless, this method needed significant computational resources for the training and deployment of DL models (Baig et al., 2022). Arbab, Q., et al. devised the Support vector machine (SVM) based classification and CNN-based feature extraction for ALL detection. This model enabled the early detection of ALL without the requirement of invasive and costly examination. However, this approach failed to utilize a large dataset to achieve generalizability (Arbab et al., 2022). Abunadi, I. and Senan, E.M., devised the ResNet-18 +SVM for the early detection of ALL. The model reduced the need for high cost and time for analyzing the blood samples. Still, this model failed to generalize to other types of leukemia samples (Abunadi and Senan, 2022).

Elhassan, T.A.M., et al. developed the Cyan Magenta Yellow, and black (CMYK) localization Moment and DL-based ALL classification. It was utilized for distinguishing different forms of WBCs. Nevertheless, this method failed to consider the parallelization and hybridization techniques for minimizing the computation time (Elhassan et al., 2022). Abrol, V., et al. devised the Modified Watershed Algorithm for the segmentation of Leukocytes. The approach achieved improved segmentation accuracy and effectively isolated and cropped specific WBCs; however, it was unable to classify individual cells accurately due to overlapping leukocytes within the same region (Abrol et al., 2023). Perveen, S., et al. Residual Convolutional Neural Network (ResNet-152)-based early detection of ALL. The model enabled the timely initiation of treatment, and it effectively classified the PBS images into healthy or cancerous. However, the real-world applicability of the model was poor (Perveen et al., 2024). Saeed, U., et al. developed the CNN-based deep learning model named DeepLeukNet for Leukemia classification. The model was used to adapt dissimilar kinds of microscopy images and detect the rare subtypes of ALL. Nonetheless, it failed to utilize the pre-trained approach for attaining multi-class classification (Saeed et al., 2024).

2.2. Challenges

The challenges faced in the existing Leukemia classification are described as follows:

- ❖ In (Atteia et al., 2022), the BO-ALLCNN was capable of minimizing an objective function and finding optimum network hyper-parameters in classification. Nevertheless, it failed to minimize noise for preserving structural details and attain clearer images.
- ❖ The SVM in (Arbab et al., 2022) was used to develop an automated detection of ALL in blood smear images with improved accuracy. However, it did not segment the blast cell to preserve the images with high resolution.
- ❖ The ResNet-18+SVM in (Abunadi and Senan, 2022) utilized deep feature maps for the extraction and classification of images with higher accuracy. Nonetheless, this model required a large data set to avoid overfitting the issue.
- ❖ In (Perveen et al., 2024), the ResNet-152 model was utilized for the early diagnosis of ALL. Moreover, it improved the overall

generalization through the leveraging of diverse models. However, the training process of the model was not faster or more stable. Hence, it failed to use the hybrid DL model to eliminate the risk of vanishing gradient.

- ❖ Leukemia impacts all aged persons like adults, children, and middle-aged persons. Hence, the detection of leukemia disorder is needed to obtain the appropriate treatment. Therefore, this work develops a hybrid DL-based approach to classify leukemia with high accuracy.

3. Proposed Deep Kronecker wide residual network -based leukemia classification

Leukemia is formed by underdeveloped WBCs, often termed as blast blood cells. The diagnosing of leukemia at an earlier stage saves the lives of individuals. However, it is complex due to the varying texture and size of the cells. This morphological variability causes more complexities for the segmentation and classification of Leukemia. To solve these issues, the DKWRN is proposed for Leukemia classification. Here, the input multi-cell blood smear images acquired from the given dataset (Acute

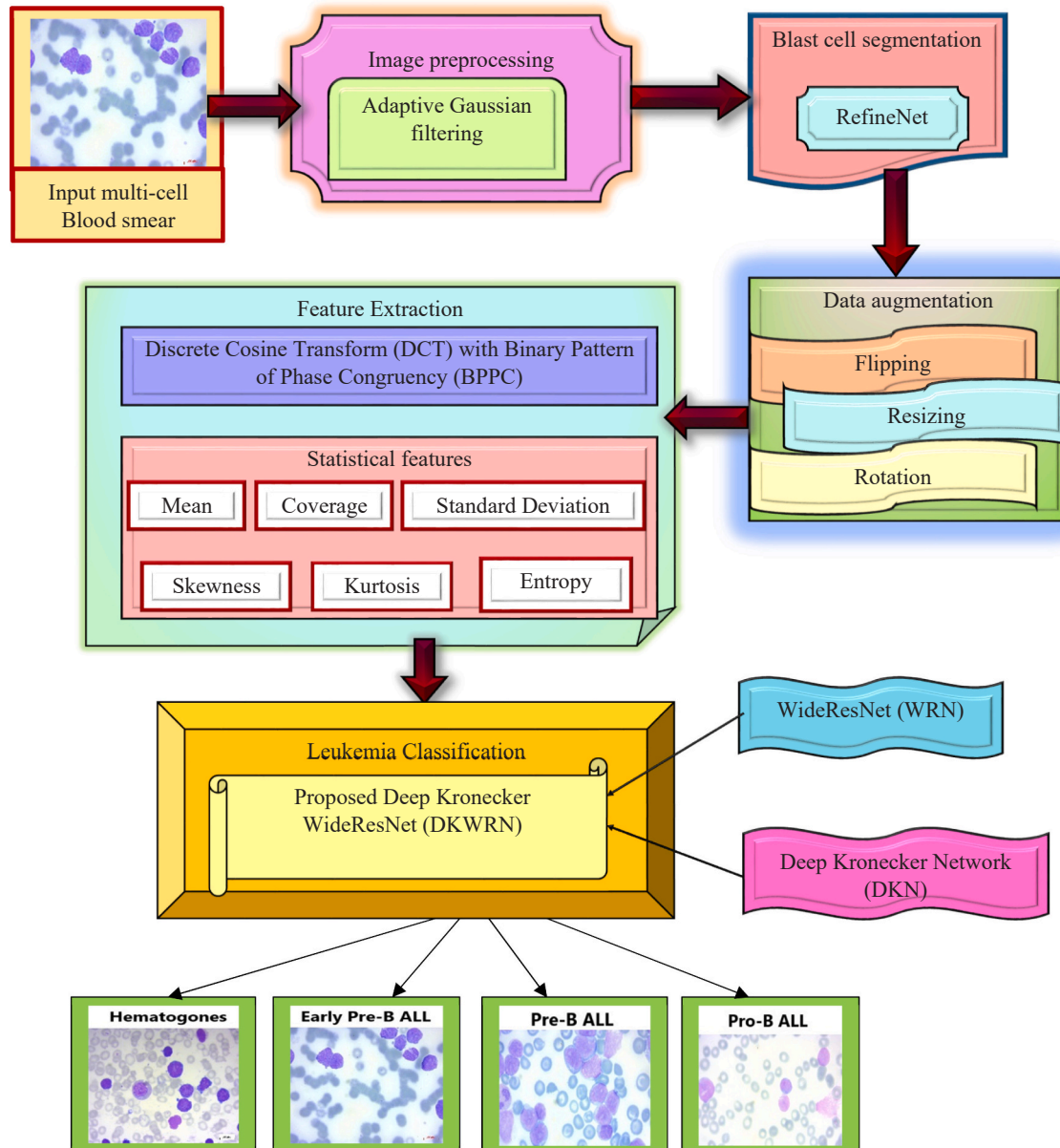


Fig. 1. Overview of the proposed DKWRN-based leukemia classification framework.

Lymphoblastic Leukemia, 2024) are subjected to image preprocessing using adaptive Gaussian filtering (Gomez, 2000). After that, Blast cell segmentation is done by RefineNet (Lin et al., 2017). Then, data augmentation (Khalifa et al., 2022) such as flipping, resizing, and rotation is carried out. Following this, features like DCT (Khayam, 2003) with BPPC (Da Fonseca et al., 2024), and statistical features (Khairandish et al., 2022) like mean, coverage, kurtosis, entropy, skewness, and standard deviation are extracted from the augmented image. The proposed DKWRN is used to classify leukemia into early Pre-B, Pre-B, Pro-B, and Hematogones by the utilization of DKWRN. Fig. 1 depicts the block diagram for Leukemia detection using the proposed DKWRN.

3.1. Image Acquisition

The input image is collected from the ALL image dataset (Acute Lymphoblastic Leukemia, 2024). It includes 3256 counts of PBS, which are gathered from 89 patients. This dataset is categorized into malignant, and benign classes, in which the benign includes hematogones and the malignant specifies the ALL with three subtypes such as Pro-B ALL, Pre-B, and early Pre-B. The overall image with the dataset is represented by,

$$D = \{D_1, D_2, \dots, D_k, \dots, D_z\} \quad (1)$$

There are z counts of images included in the dataset, in which the k^{th} image is symbolized by D_k .

3.2. Image Preprocessing using Adaptive Gaussian Filter

Preprocessing is the initial step for Leukemia classification. The image obtained from the dataset contain a nonuniform resolution. In this context, the preprocessing stage plays a crucial role in preparing the image data for effective analysis. It involves resizing the images to a uniform dimension, which ensures consistency across the dataset and facilitates efficient model training. Additionally, standardization of images is applied to normalize pixel values, thereby improving convergence during learning. Beyond these steps, preprocessing also includes noise reduction methods that help eliminate unwanted artifacts and enhance image clarity. Furthermore, it aids in suppressing irrelevant or redundant visual information, allowing the model to focus more accurately on the meaningful features essential for classification. Here, the adaptive Gaussian filter is used for image preprocessing. The adaptive Gaussian filter (Gomez, 2000) minimizes the noise while conserving the significant features, and it changes the behavior as on the characteristics based on the local image. Furthermore, the filter adapts the Gaussian function and standard deviation as per the local region. It creates a stronger smoothing of uniform regions while preserving the detail of complex areas. By varying the degree of smoothing using local gradients, the edge regions are preserved. Consider, the input image D_k with the Gaussian Kernel B_α , where $\alpha = \alpha_1, \dots, \alpha_y$. It is essential to select the relevant scale (α) for every pair (w, x) . Hence, the ideal coding is necessary to describe the parameters. The preprocessed image H_k obtained from the adaptive Gaussian filter is formulated as,

$$H_k(w, x) = D_k(w, x) + \beta_\alpha(w, x) \quad (2)$$

From Eq. (2), the preprocessed image H_k is obtained and the residual is denoted by β_α .

3.3. Blast cell segmentation using RefineNet

Blast cell segmentation is the identification and isolation of blast cells or immature WBC from the bone marrow or blood smear samples. The image H_k is fed to the blast cell segmentation. Moreover, the segmentation of blast cells is needed to diagnose and monitor the hematological illnesses, especially in leukemias. The identification of blast

cells indicates numerous classes of leukemia. Here, the segmentation of blast blood cells is performed using the RefineNet, and the segmented outcome is indicated by J_k . The structural description of RefineNet is described below:

3.3.1. Structure of RefineNet

RefineNet (Lin et al., 2017) is a type of DL which is utilized for high-resolution semantic segmentation. Moreover, it integrates the refined segmentation into the multi-scale feature aggregation. The structure of RefineNet is given in Fig. 2. The primary component of every RefineNet block fine-tunes the weights in a pre-trained ResNet. Every input channel is successively fed to residual convolution units (RCU), a multi-resolution fusion unit, and chained residual pooling (CRP).

a) RCU: RCU is a simplified form of a convolution (Conv) unit, in which the Batch normalization (BN) is eliminated. Moreover, the outcome of RefineNet is passed to the multi-resolution fusion.

b) Multi-resolution fusion: It fuses the inputs to generate the feature map with high resolution. It carries out the convolutions for input adaptation and forms feature maps with similar sizes. It upsamples the feature maps to a higher resolution. At last, the feature maps are merged using the summation operation. Moreover, the adapted input rescales the feature in a different manner, getting the successive sum combination.

c) CRP: The feature map from the multi-resolution fusion unit is sent through the CRP block, where the background details from the image are extracted. It efficiently pools the features from several window sizes and combines them to form a learnable weight. CRP includes several pooling units, in which every block includes one Conv and one max-pooling unit. Each pooling block receives the outcome of the preceding one. The feature maps generated by the pooling stage are integrated with the original input maps. Subsequent convolutional operations within the block act as a weighted aggregation mechanism, facilitating the fusion process through summation. In the training phase, the Conv layer detects the significance of the pooling block.

d) Output convolutions: Every RefineNet block is connected to another residual Conv unit. It creates a series of three RCUs in every block. Hence, it performs the nonlinearity function on multi-path fused feature maps and creates the segmented outcome J_k .

3.4. Augmentation of the segmented image

Image augmentation (Khalifa et al., 2022) is needed for increasing the generalization, and robustness of the image. Hence, it enables the images to be highly robust in various conditions and patient discrepancies. Moreover, the augmentation creates variations such as flips, rotations, and brightness resizing. The blast cell segmented image J_k is fed to the augmentation. The image augmentation methods, like flipping, rotations, and resizing are described below:

3.4.1. Flipping

Flipping (Khalifa et al., 2022) is needed to reverse the pixels of an image along with horizontal or vertical, or both axes. Moreover, it manipulates the pixel coordinates of the image matrix. Here, the image J_k has the coordinates (a, b) . The flipping is described as follows:

3.4.1.1. Vertical flipping. In vertical flipping, the vertical, and horizontal axis of the image is flipped based on the bottom and top of the up-down side. The current coordinates pixel after vertical flipping is given by,

$$\begin{bmatrix} \psi_a^{VF} \\ \psi_b^{VF} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} \quad (3)$$

Here, the pair of coordinates are represented by a and b . Moreover,

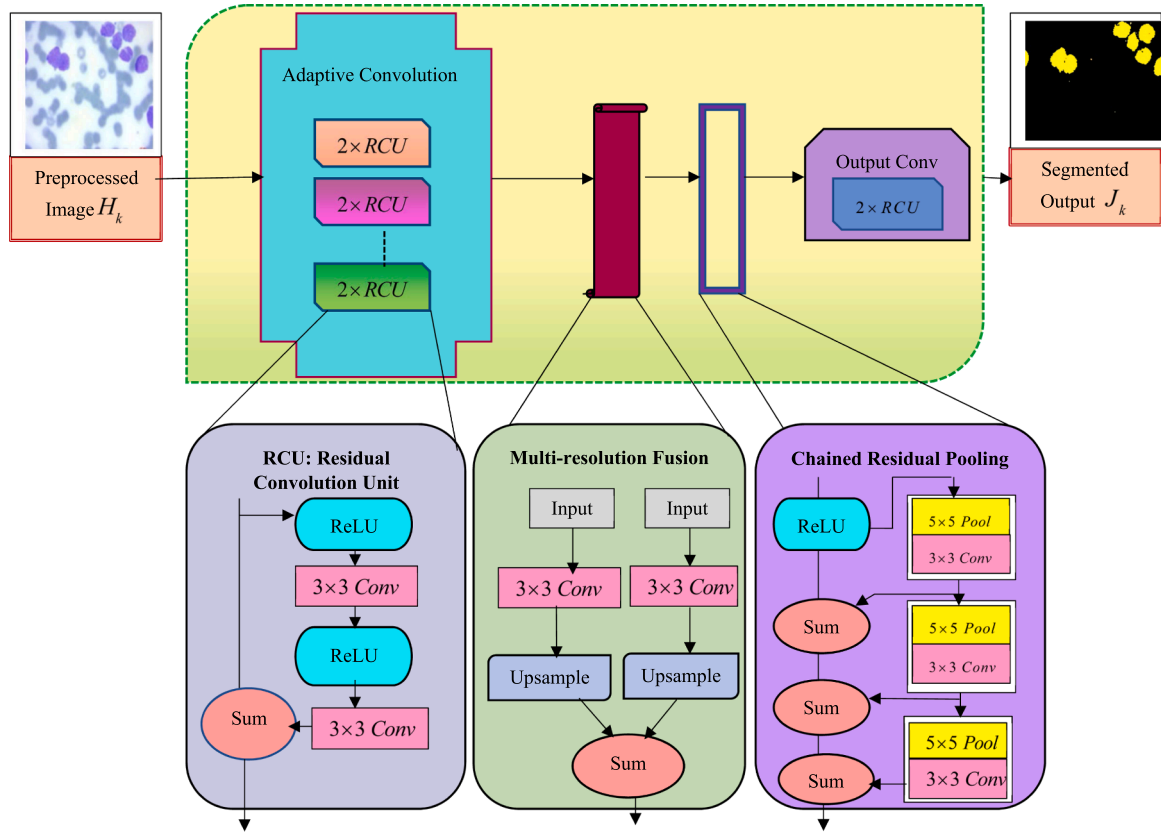


Fig. 2. Schematic representation of the RefineNet architecture.

$\begin{bmatrix} \psi_a^{VF} \\ \psi_b^{VF} \end{bmatrix}$ specifies the coordinates specify after vertical flipping.

3.4.1.2 Horizontal flipping

In horizontal flipping, the image's right, and left sides are interchanged in a horizontal manner. After performing the horizontal flipping, the coordinates are expressed as,

$$\begin{bmatrix} \psi_a^{HF} \\ \psi_b^{HF} \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} \quad (4)$$

where, the coordinates after horizontal flipping are specified as $\begin{bmatrix} \psi_a^{HF} \\ \psi_b^{HF} \end{bmatrix}$.

3.4.1.3 Vertical- Horizontal flipping

This type of filtering is a merging of both horizontal and vertical and flipping, in which the image is flipped both top-to-bottom and left-to-right by the angle of 180°.

$$\begin{bmatrix} \psi_a^{VHF} \\ \psi_b^{VHF} \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} \quad (5)$$

where, $\begin{bmatrix} \psi_a^{VHF} \\ \psi_b^{VHF} \end{bmatrix}$ implies the coordinates after vertical-horizontal flipping. Moreover, the overall flipping-based augmentation is indicated by the term L_k^{FL} .

3.4.2. Rotation

In rotation (Khalifa et al., 2022), the image is rotated by a certain angle along with the right and left sides. Moreover, the rotating angle is varied from 1 to 360°. Here, rotation is expressed by,

$$\begin{bmatrix} \psi_a^{RO} \\ \psi_b^{RO} \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} \quad (6)$$

where, the rotation is indicated by L_k^{RO} .

3.4.3. Resizing

The size of the image is changed to create new variations, and it modifies the image's dimensionality while increasing the diversity. Moreover, the resized image is denoted as L_k^{RE} .

Furthermore, the overall augmented image L_k is expressed as follows,

$$L_k = \{L_k^{FL}, L_k^{RO}, L_k^{RE}\} \quad (7)$$

where, L_k^{FL} , L_k^{RO} , and L_k^{RE} represents the flipped, rotated, and resized images.

3.5. Extraction of features from the augmented image

Extracting features is a crucial process that reduces the dataset's dimensionality while retaining the most informative and relevant attributes needed for accurate analysis. The extraction of significant features offers meaningful illustrations of the image. In this stage, the input image L_k serves as the basis for deriving features, where BPPC, combined with DCT and statistical descriptors, are systematically extracted.

3.5.1. BPPC with DCT Feature

Phase congruence (PC) (Da Fonseca et al., 2024) features are used as an important pattern descriptor because they provide an independent signal magnitude for attaining contrast changes, and insensitive image illumination. Unlike edge detectors, it detects abrupt variations in intensity. Moreover, the PC offered a strong ability for recognizing and describing the images, as well as separating samples for different classes.

For computing the PC, the spatial frequency details of the image are required, which is typically accomplished via the utilization of a wavelet transform. The image L_k is applied to PC extraction, which represents every pixel point of the image. Moreover, even and odd symmetrical wavelets for the scale d are specified by the terms I_d^E and I_d^O . The response for every pair of quadrature filters creates the response vector as follows,

$$[E_d(c), O_d(c)] = [L_k(c) * I_d^E(c) * I_d^O(c)] \quad (8)$$

where, $E_d(c)$, and $O_d(c)$ specifies the even, and odd response. Furthermore, the PC is formulated by the ratio of normalized local energy $[e(c)]$ to the amplitude factor $[G_d(c)]$.

$$PC(c) = \frac{e(c)}{\sum_d G_d(c)} \quad (9)$$

By the consideration of noise, the PC becomes,

$$PC(c) = \frac{e(c) - K}{\sum_d G_d(c) + M} \quad (10)$$

where, K implies the noise threshold, and M specifies the slight offset.

The binary pattern of PC indicates the variation of Local Binary Patterns (LBP). Moreover, the BPPC computes the phase congruency images of every orientation, and the resultant feature set is attained through the concatenation of every BPPC image histogram. It effectively captures structural and textural details, such as edges, cell boundaries, and chromatin patterns in leukemic cells, which are vital for distinguishing between ALL subtypes. The feature obtained from the BPPC is indicated by F_{1k}^{BPPC} .

After the extraction of BPPC, DCT (Khayam, 2003) is applied to attain the BPPC with DCT features. Moreover, DCT includes the sampled cosine operators and basis vectors. DCT minimizes redundant data and minimizes the burden of computational complexity. Here, the BPPC feature $F_k^{BPPC}(s_1, s_2)$ with dimension $[S_1 \times S_2]$ is fed to DCT to obtain $F_k^{DCT}(u_1, u_2)$ as,

$$F_k^{DCT}(u_1, u_2) = h(u_1)h(u_2) \sum_{s_1=0}^{S_1-1} \sum_{s_2=0}^{S_2-1} F_k^{BPPC}(s_1, s_2) \cos\left(\frac{\pi(2s_1+1)u_1}{2S_1}\right) \cos\left(\frac{\pi(2s_2+1)u_2}{2S_2}\right), 0 \leq u_1 \leq S_1-1; 0 \leq u_2 \leq S_2-1 \quad (11)$$

From F_k^{DCT} bands like Higher Lows (HL), Lower Lows (LL), and Lower Highs (LH) are obtained. Furthermore, the feature vector attained through F_k^{DCT} is represented by the term F_{1k} .

3.5.2. Statistical features

The augmented image L_k is also applied to the statistical features extraction, where the features (Khairandish et al., 2022; Cristin et al., 2021) like mean, variance, kurtosis, skewness, entropy, and standard deviation are extracted. The following section describes the mathematical formulation for statistical features:

a) Mean: Mean (Khairandish et al., 2022) is defined as the brightness of the image as per the significant count of bright pixels. Additionally, the mean value is determined by averaging the intensity across all pixel elements, calculated as the sum of pixel values divided by their total number.

$$Mean = \frac{1}{\sigma \times v} \sum_{\xi=0}^{\sigma-1} \sum_{\lambda=0}^{v-1} \Re(\xi, \lambda) \quad (12)$$

Here, the highest counting of pixels is denoted by $\sigma \times v$, and $\Re(\xi, \lambda)$ shows the pixel of the image on the axis (ξ, λ) . Furthermore, the mean is indicated by F_{2k}^{Me} specifies the mean.

b) Coverage: The coverage feature (Cristin et al., 2021) is computed

by the shape of the image, and it is needed to categorize the image. Furthermore, the coverage feature is computed by analyzing both the segmented pixel regions and the total pixel area of the image, allowing for a comprehensive assessment of spatial distribution. Here, the coverage is signified as F_{2k}^{Co} .

c) Skewness: Skewness (Khairandish et al., 2022) describes the excessive condition of a database or the distribution about the mean. It is simple to compute and adds minimal computational burden while enhancing the discriminative capability of the feature set used for classification. The skewness is denoted by,

$$Skewness = \frac{h_3}{\gamma^3} \quad (13)$$

The term γ^3 shows the standard deviation, and the third moment of the mean is represented by h_3 .

$$h_3 = \frac{1}{\sigma \times v} \sum_{\xi=0}^{\sigma-1} \sum_{\lambda=0}^{v-1} (\Re(\xi, \lambda) - Mean)^3 \quad (14)$$

The skewness is denoted by F_{2k}^{SK} .

d) Kurtosis: Kurtosis (Khairandish et al., 2022) is a statistical measure for describing the distribution of data points. It is computationally efficient to calculate and adds discriminative value to the overall feature set without significantly increasing processing time. Moreover, the Kurtosis is formulated by,

$$Kurtosis = \frac{h_4}{\gamma^4} \quad (15)$$

The fourth moment about the mean is symbolized as h_4 .

$$h_4 = \frac{1}{\sigma \times v} \sum_{\xi=0}^{\sigma-1} \sum_{\lambda=0}^{v-1} (\Re(\xi, \lambda) - Mean)^4 \quad (16)$$

The Kurtosis is indicated by the term F_{2k}^{Ku} .

e) Entropy: Entropy (Khairandish et al., 2022) is used for identifying the texture data of the image, where $og_2 \Re(\xi, \lambda)$ computes the textural description of the image. It effectively captures differences between normal and leukemic cells by quantifying textural randomness. Cells with disordered internal structures typically yield higher entropy values compared to healthy ones.

$$Entropy = \sum_{\xi=0}^{\sigma-1} \sum_{\lambda=0}^{v-1} \Re(\xi, \lambda) \log_2 \Re(\xi, \lambda) \quad (17)$$

The term F_{2k}^{En} indicates entropy.

f) Standard Deviation: Standard deviation (Khairandish et al., 2022) is used for measuring the quantity of dispersion of individual data from the mean value. Here, the standard deviation is expressed by,

$$Standard\ Deviation\ \gamma^3 = \sqrt{\frac{1}{\sigma \times v} \sum_{\xi=0}^{\sigma-1} \sum_{\lambda=0}^{v-1} (\Re(\xi, \lambda) - mean)^2} \quad (18)$$

The standard deviation feature is indicated F_{2k}^{Sd} .

Hence, the overall statistical features are given by,

$$F_{2k} = \{F_{2k}^{Me}, F_{2k}^{Co}, F_{2k}^{SK}, F_{2k}^{Ku}, F_{2k}^{En}, F_{2k}^{Sd}\} \quad (19)$$

Furthermore, the feature vector extracted from augmented image L_k are denoted by F_k .

$$F_k = F_{1k} || F_{2k} \quad (20)$$

where, F_{1k} shows the feature vector attained from BPPC with DCT feature, and the feature vector F_{2k} is got from the statistical features.

In the proposed DKWRN framework, the feature fusion of BPPC, DCT, and statistical features is performed through a feature-level concatenation strategy. Specifically, the extracted features from each descriptor are first normalized to ensure uniform scale, and then

concatenated into a single comprehensive feature vector before being fed into the DKWRN for classification. This approach allows the model to jointly learn complementary spatial, frequency, and statistical representations of leukemic cells, thereby enhancing discriminative power and overall classification accuracy.

3.6. Leukemia classification using proposed DKWRN

The classification of leukemia is needed to offer effective treatment for curing the Leukemia disorder. Here, the DKWRN is used to classify the Leukemia, in which the integration of WRN (Zagoruyko and Komodakis, 2016; Zhang et al., 2021) and DKN (Feng and Yang, 2022) develops the DKWRN. Here, WRN contributes by increasing the width of residual blocks, which allows the model to learn richer and more diverse feature maps without significantly increasing depth, thereby improving training efficiency and mitigating the vanishing gradient problem. Simultaneously, the inclusion of DKN introduces Kronecker convolution operations, which enable compact and structured parameterization of convolutional layers, leading to improved modeling of spatial correlations and reduced parameter redundancy. This synergistic integration allows DKWRN to capture both high-level abstract features and detailed local structures of blast cells more effectively, which is crucial for distinguishing between closely related leukemia subtypes. Fig. 3 illustrates the schematic view of DKWRN. Initially, the preprocessed image H_k is given to the WRN for getting the output U_1 . Following that, the output of WRN(U_1) and the feature vector F_k are fed to the Wide Kronecker layer, in which the regression computes the similarity among the target and extracted features for getting the output U_2 . The combination of pre-processed image H_k , and U_2 are named by a single term A_k is fed to the DKN. In the DKN model, the classified outcome U_k is obtained. Moreover, the hybrid network is tuned using Adam optimization.

Table 1 shows the architectural comparison of DKWRN. Here, the number of layers in DKWRN is 47 layers.

3.6.1. WRN

WRN (Atteia et al., 2022; Zhang et al., 2021) is a variety of ResNet, which is used for widening the network layers and increasing the count of feature maps. Here, the preprocessed image H_k is fed to WRN, which contains the residual blocks, in which every block has skip or shortcut links. These links enable the network to transmit information from one layer to another. Additionally, skip connections are utilized to facilitate the learning of identity mappings, effectively mitigating challenges related to vanishing gradients during deep network training. It effectively stabilizes depth with wider layers for diminishing the training period. In the residual block, the representational power is increased by

Table 1
Architectural table of DKWRN.

Architecture/ Stage	Layer Type	Kernel Size	Filters/ Units	Output Shape
Input	Input	—	—	(32, 32, 3)
WRN Path	Conv2D	3×3	16	(32, 32, 16)
	BatchNorm + ReLU	—	—	(32, 32, 16)
WRN Block Group 1	$2 \times \text{Conv2D} (\times 2 \text{ blocks})$	3×3	32	(32, 32, 32)
	Shortcut Conv	1×1	32	(32, 32, 32)
WRN Block Group 2	$2 \times \text{Conv2D} (\times 2 \text{ blocks})$	3×3	64	(16, 16, 64)
	Shortcut Conv	1×1	64	(16, 16, 64)
WRN Block Group 3	$2 \times \text{Conv2D} (\times 2 \text{ blocks})$	3×3	128	(8, 8, 128)
	Shortcut Conv	1×1	128	(8, 8, 128)
WRN Output	BN + ReLU → AvgPooling	—	—	(1, 1, 128)
	Flatten → Dense	—	128	(128)
Kronecker Path	Conv2D (1)	3×3	32	(32, 32, 32)
	Conv2D (2)	3×3	32	(32, 32, 32)
	Multiply	—	—	(32, 32, 32)
	Flatten → Dense	—	128	(128)
Fusion	Add or Concatenate	—	—	(128) or (256)
Classifier	Dense	—	4	(4)

adding more Conv layers at every block, adding several feature planes, and enhancing the size of the filter. Furthermore, the widening factor is utilized to measure the width of the Conv layers, and the deepening factor computes the quantity of Conv in a block. Moreover, the structure of WRN is given in Fig. 4.

a) Conv layer in residual block: The primary Conv layer is followed by 3 sets of residual blocks, which are followed by average pooling and softmax layers. The input is sent to 3×3 Conv layer with a certain count of filters for minimizing the size of the image.

b) Width for the Residual blocks: In WRN, the block deepening parameter i negatively impacts the model while considering a similar count of parameters. Hence, it is required to set different counts of i and count of blocks j . As a result, the computational complexity and the number of parameters are quadratic for the widening parameter L .

c) Dropout in the Residual block: Residual architectures

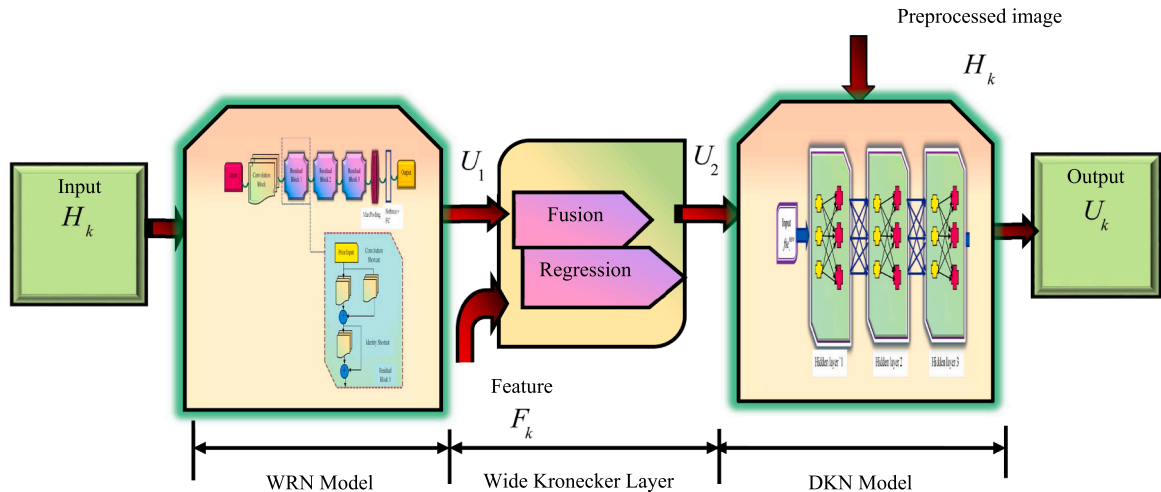


Fig. 3. Schematic representation of the proposed DKWRN architecture.

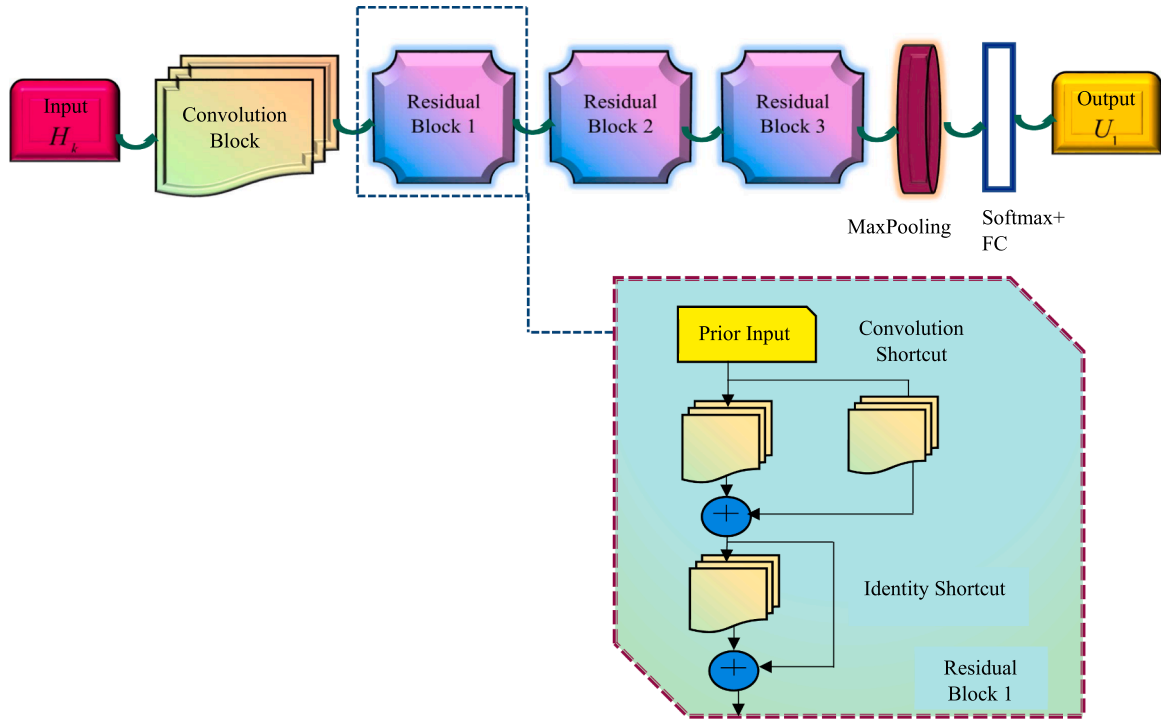


Fig. 4. Schematic view of WRN.

incorporate batch normalization (BN) to enhance regularization, yet this often demands extensive data augmentation. To mitigate this challenge, each convolutional residual block is supplemented with a dropout layer to promote training efficiency and reduce overfitting. Moreover, it controls the feature reuse and enforces the learning of diverse residual blocks.

d) Skip Connection: The convolutional layer and identity shortcut are linked to bridge the input and output while preserving the original data pathway unchanged.

e) Pooling Layer: The feature map is passed into the pooling layer, where each map undergoes spatial compression, reducing it to a singular representative value per channel.

f) Fully Connected Layer: The aggregated features from the pooling operation are forwarded to a dense layer, where the number of output neurons corresponds directly to the total classification categories.

g) Softmax Activation: The final result is fed into a softmax activation layer, which converts the raw outputs into a probability distribution across the target classes to identify the most likely prediction. Using the Residual block with identity mapping, the output of the WRN (U_1) is obtained using the expression below,

$$U_1 = H_K + \text{fun}(H_K, Q_r) \quad (21)$$

where, Q_r implies the parameter of the block, H_K specifies the input, and fun implies the residual function.

3.6.2. Wide Kronecker Layer

The output of WRN (U_1), and the feature vector F_k are given to the In the Wide Kronecker layer. Fusion and regression are the crucial functions to create U_2 . Here, the Wide Kronecker Layer fusion is designed to enhance feature representation by capturing both local and global interactions across multiple receptive fields, which is especially critical in differentiating subtle morphological variations among leukemic cells. This fusion effectively increases the model's capacity without significantly increasing computational complexity. Similarly, the regression update rules are grounded in minimizing classification error while ensuring stable convergence during training. These rules were adopted after empirical evaluations demonstrated improved accuracy and

robustness. Here, the outcome at the interval η is given as,

$$\mu = \sum_{g=1}^w (F_k)_g * \varpi_g \quad (22)$$

Here, F_k implies the feature vector, and g represents the counts of features. The weight is indicated by the term ϖ and the output at the period η is portrayed as μ .

Using the Fractional Calculus (FC) (Bhaladhare and Jinwala, 2014), the updates expression at the iteration ($\eta + 1$) is given as,

$$\mu(\eta + 1) = \varsigma \cdot \mu(\eta) + \frac{1}{2} \varsigma \cdot \mu(\eta - 1) \quad (23)$$

Consider $\mu(\eta + 1) = U_2$ and $\mu(\eta - 1) = U_1$. Hence Eq. (23) becomes,

$$U_2 = \varsigma \cdot \mu(\eta) + \frac{1}{2} \varsigma \cdot U_1 \quad (24)$$

Substitute Eq. (24) in Eq. (22),

$$U_2 = \varsigma \sum_{g=1}^w (F_k)_g * \varpi_g + \frac{1}{2} U_1 \quad (25)$$

$$U_2 = \varsigma \sum_{g=1}^w (F_k)_g * \varpi_g + \frac{1}{2} [H_K + \text{fun}(H_K, Q_r)] \quad (26)$$

where, the outcome of the DKWRN layer is indicated by U_2 .

3.6.3. DKN

DKN (Feng and Yang, 2022) is a form of neural network structure, which controls the Kronecker product. Moreover, Kronecker creates the block matrix. The input, hidden, and output layers are the part of DKN. After the estimation of the Kronecker product, the resultant matrix is reshaped and flattened. The DKN solves the complex relation of features without the utilization of an excessive count of parameters. The architecture of DKN is portrayed in Fig. 5, in which the combination of features F_k and the outcome of the DKWRN layer (U_2) are specified by a single term A_k . Here A_k is fed to the DKN, and it yields the following

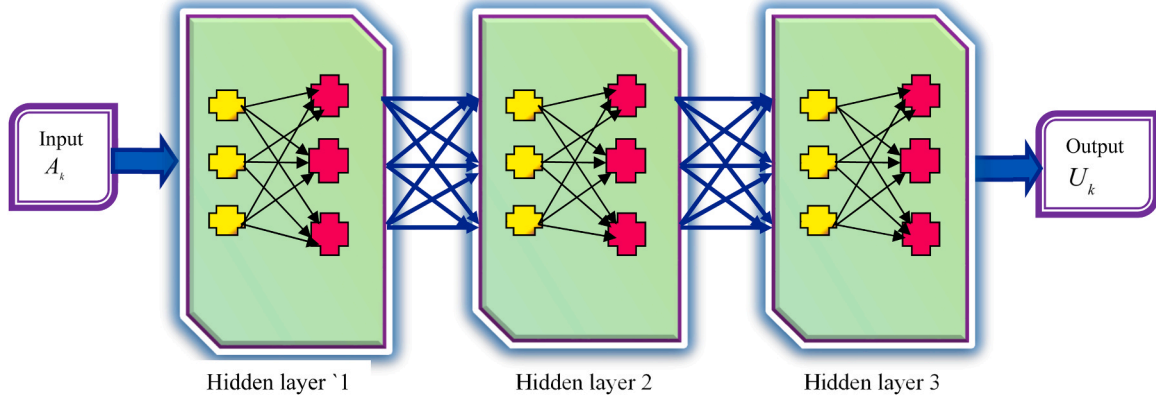


Fig. 5. Schematic model of DKN.

outcome,

$$U_k = \delta(n_q) \text{Exp} \{ n_q \langle (U_2)_q, R \rangle - \gamma \langle (U_2)_q, R \rangle \} \quad (27)$$

$$U_k = \delta(n_q) \text{Exp} \left\{ \begin{aligned} & n_q \left\langle \left(\left[\sum_{g=1}^W (F_k)_g * \varpi_g + \frac{1}{2} [H_K + \text{fun}(H_K, Q_r)] \right] \right)_q, R \right\rangle - \\ & \gamma \left\langle \left(\left[\sum_{g=1}^W (F_k)_g * \varpi_g + \frac{1}{2} [H_K + \text{fun}(H_K, Q_r)] \right] \right)_q, R \right\rangle \end{aligned} \right\} \quad (28)$$

The output of DKN is portrayed by U_k . Furthermore, $\delta(\cdot)$ and $\gamma(\cdot)$ illustrate the known univariate operation. The expression for certain known link operations $e(\cdot)$ is given as,

$$e(\lambda(n_q)) = \langle (U_2)_q, R \rangle \quad (29)$$

The decomposition of the Kronecker product $N(\geq 2)$ using the rank P , and the coefficient R is expressed as,

$$R = \sum_{p=1}^P V_t^p \otimes V_{t-1}^p \otimes \dots \otimes V_1^p \quad (30)$$

The term $p = 1, \dots, P$ indicates the Kronecker component and $t = 1, \dots, T$ denotes the known matrix. Moreover, $\lambda = \Pi_{u=1}^L \lambda_u$ and $\mathfrak{F} = \Pi_{u=1}^U \mathfrak{F}_u$ is attained through the Kronecker product. The simplified expression is given as,

$$V_t \otimes V_{t-1} \otimes \dots \otimes V_{t'} = \sum_{t=t'}^{t'} V_\mu \quad (31)$$

In each matrix, the decomposition is denoted as $R = \sum_{p=1}^P \otimes_{t=T}^1 V_t^p$.

4. Results and discussion

This section details the performance assessment of the proposed DKWRN model in classifying Leukemia. Additionally, this section outlines the employed software tools, details the dataset utilized, evaluates the performance of the DKWRN model for Leukemia classification, and presents a comparative analysis highlighting its effectiveness against alternative approaches.

4.1. Experiment setup

The DKWRN-based Leukemia classification is implemented using the PYTHON tool. Moreover, the hyperparameter tuning is performed based on the method called grid search. Here, Grid search is a method for tuning hyperparameters that systematically examines all possible combinations within a defined range, aiming to identify the optimal configuration for a machine learning model. Moreover, it offers a

systematic and thorough approach to hyperparameter tuning by evaluating all possible combinations of parameters within a predefined range, ensuring no potential combination is overlooked. Moreover, the deterministic nature of grid search guarantees reproducibility, allowing researchers to consistently replicate results. By identifying the optimal hyperparameters, it improves the performance and accuracy of the model, enabling better results in predictive tasks. Despite being computationally intensive, especially for large parameter spaces, grid search remains a reliable and effective method for achieving well-tuned models in smaller or manageable problems. The experimental parameters are shown in Table 2.

4.2. Dataset description

ALL image datasets ([Acute Lymphoblastic Leukemia, 2024](#)) are utilized in this dataset. It includes 3256 PBS collected from 89 patients suspected of having ALL. Blood samples were carefully prepared and stained by experienced laboratory professionals. The images are categorized into two main classes: benign and malignant. The benign class includes hematogones, while the malignant class represents ALL cases, further divided into three subtypes of lymphoblasts: Early Pre-B, Pre-B, and Pro-B ALL. Here, the dataset is used 80 % for training, 10 % for testing, and 10 % for validation. Leukemia-Classification Computer Vision Dataset ([Leukemia-Classification, .](#)) is a curated collection of single-cell images derived from peripheral blood smears of patients diagnosed with ALL. It contains approximately 3242 images, 4 original classes and each is sized at 224×224 pixels.

Table 2

Experimental parameters.

	Parameters	Values
Hyperparameters	epochs	100
	batch size	64
	Learning rate	0.001
	dropout rate	0.5
	optimizer	adam
	loss	categorical_crossentropy
	activation	relu
	Weight decay	0.0005
	number of layers	47 layers
	size of kernel	(3,3), (1,1)
	WRN widening factor	2
	filters	[32,64,128]
	stride	2
Hardware specifications	RAM	16 GB
	GPU	2 GB
	ROM	More than 100 GB
	CPU	2.4 GHz

4.3. Experimental outcomes

Figs. 6a) through 6j) collectively illustrate the complete processing pipeline for Image-1, beginning with the raw PBS image, which is presented in Fig. 6a, highlighting the natural variability and noise in the sample. The subsequent preprocessing step in Fig. 6b employs adaptive Gaussian filtering to reduce noise and enhance cellular structures, followed by RefineNet-based segmentation, which is shown in Fig. 6c, that isolates the blast cell, ensuring that only relevant regions are analyzed. The augmentation steps, including flipping in Fig. 6d, resizing the Fig. 6e, and rotation in Fig. 6f, expand the dataset diversity and improve the model's robustness to variations in orientation and scale. Feature extraction is then performed, with BPPC features illustrated in Fig. 6g, capturing edge and texture information, and DCT features in Fig. 6h encoding complementary frequency-based patterns. These combined features feed into the DKWRN for final classification, as shown in Fig. 6i, which presents the predicted ALL subtype for the image. Finally, Fig. 6j displays the Grad-CAM visualization, highlighting the regions of the cell that the model attends during classification, thereby confirming that DKWRN focuses on biologically relevant structures such as the nucleus and cytoplasm. Here, the Grad-CAM visualizations clearly indicate that the proposed DKWRN model focuses its attention on the most discriminative regions of the leukemic cells, particularly around the nucleus and cytoplasmic boundaries, which are critical for distinguishing different subtypes of ALL. The highlighted regions in the Grad-CAM maps correspond closely with the areas identified by hematologists as diagnostically significant, demonstrating that the model effectively captures relevant morphological features rather than relying on background or irrelevant regions. Moreover, the consistency of these attention maps across multiple samples confirms that DKWRN learns stable and biologically meaningful patterns for classification.

Likewise, Figs. 7a), 7b), 7c), 7d), 7e), 7f), 7g), 7h), and 7i) depict the input, preprocessed, segmented, flipped, resized, rotated, BPPC extracted, DCT feature extracted, and classified outcome for image-2. The Grad-CAM output of image-2 is presented in Fig. 7j).

4.4. Performance Computing measures

The measurements used for assessing the DKWRN-based Leukemia classification are given as follows:

4.4.1. Accuracy

Accuracy (Da Fonseca et al., 2024) is used for detecting the percentage of exact detection of the disease. Moreover, the accuracy is formulated by,

$$Accuracy = \frac{TR^+ + FA^-}{TR^+ + FA^+ + TR^- + FA^-} \quad (32)$$

The true positive and false positive are specified by TR^+ and FA^+ , and the term TR^- and FA^- implies the true negative and false negative.

4.4.2. Recall (TPR)

Recall (Da Fonseca et al., 2024) represents the portion of the whole counting of positively detected cases to the count of exactly identified positives. Moreover, the recall is given by,

$$TPR = \frac{TR^+}{TR^+ + FA^-} \quad (33)$$

4.4.3. TNR

TNR (Da Fonseca et al., 2024) specifies the percentage of the entire negatively identified cases to the number of precise negatives. The TNR is given as,

$$TNR = \frac{TR^-}{TR^- + FA^+} \quad (34)$$

4.4.4. Precision

Precision is a performance metric used in classification tasks to measure the accuracy of positive predictions.

$$Precision = \frac{TP}{TP + FP} \quad (35)$$

4.4.5. F1-score

The F1-score is a performance metric that combines both precision and recall into a single value, providing a balanced measure of a model's accuracy, especially in cases of imbalanced datasets.

$$F1 - score = 2 \times \frac{precision \times recall}{precision + recall} \quad (36)$$

4.5. Performance evaluation

This part represents the performance computation of DKWRN-based Leukemia classification regarding the changing of training data and K-Fold. Moreover, the performance assessment is illustrated as follows:

4.5.1. Analysis based on training data

Fig. 8 presents the training-based performance evaluation of the DKWRN model for Leukemia classification, assessed across multiple epochs specifically at intervals of 20, 40, 60, 80, and 100 to analyze its learning progression and accuracy trend. The analysis of DKWRN in terms of accuracy is exhibited in Fig. 8a). For the training data of 90 %, the DKWRN attained an accuracy of 86.29 %, 87.45 %, 88.89 %, 89.14 %, and 91.64 % for the epochs 20–100. Fig. 8b) demonstrates how the TNR has been quantitatively assessed within the framework of the proposed model. Using the training data= 90 %, the TNR of DKWRN under the 20–100 epochs are 85.61 %, 86.49 %, 87.37 %, 88.47 %, and 90.59 %, respectively. The estimation with respect to recall is deliberated in Fig. 8c). Considering 90 % of training data, the recall of DKWRN for epochs 20 is 84.15 % 40 is 85.26 % 60 is 86.27 % 80 is 87.17 and 100 is 89.27 %. The analysis of DKWRN in terms of precision is exhibited in Fig. 8d). For the training data of 90 %, the DKWRN attained a precision of 85.93 %, 87.17 %, 88.51 %, 88.65 %, and 91.21 % for the epochs 20–100. Fig. 8e) illustrates the estimation related to F1-score. Using the training data= 90 %, the F1-score of DKWRN under the 20–100 epochs are 85.03 %, 86.21 %, 87.38 %, 87.91 %, and 90.23 %, respectively.

4.5.2. Analysis based on K-Fold

Fig. 9 portrays the performance computation of DKWRN-based Leukemia classification with respect to with K-Fold. The analysis of DKWRN in terms of accuracy is shown in Fig. 9a). Under the K-Fold of 9, the DKWRN obtained the accuracy of 87.32 %, 88.90 %, 89.19 %, 90.13 %, and 92.26 % for the epochs varied from 20 to 100. Fig. 9b) illustrates the estimation related to recall. For the K-Fold= 9, the TNR of DKWRN for the epochs varied from 20 to 100 are 85.02 %, 87.30 %, 88.71 %, 89.07 %, and 91.38 %, respectively. In addition, the evaluation concerning recall is deliberated in Fig. 9c). The recall of 84.79 % 86.47 %, 87.24 %, 88.36 %, and 90.42 % are achieved by the DKWRN at 20th, 40th, 60th, 80th, and 100th with respect to the K-Fold of 9. Fig. 9d) presents the evaluation of precision metrics derived from the DKWRN framework. Under the K-Fold of 9, the DKWRN obtained the precision of 86.89 %, 88.73 %, 88.63 %, 89.34 %, and 91.72 % for the epochs varied from 20 to 100. The graphical output in Fig. 9e) reflects the performance assessment based on the F1-score parameter. For the K-Fold= 9, the F1-score of DKWRN for the epochs varied from 20 to 100 are 85.83 %, 87.59 %, 87.93 %, 88.85 %, and 91.07 %, respectively.

4.6. Comparative techniques

The methods like BO-ALLCNN (Atteia et al., 2022), SVM (Arbab et al., 2022), ResNet-18 +SVM (Abunadi and Senan, 2022), ViT-CNN

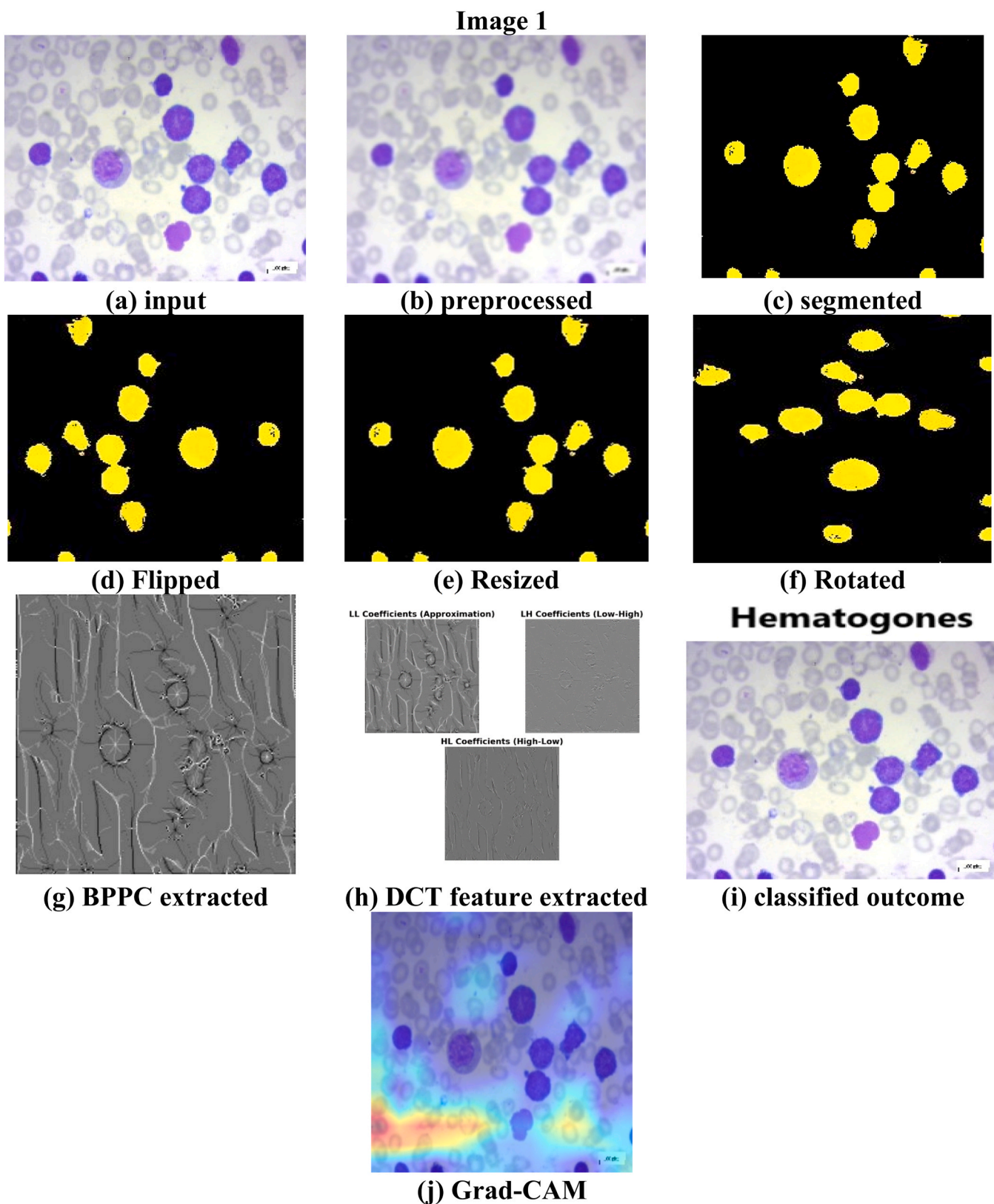


Fig. 6. Experimental results of sample images 1, 6 a) input, 6 b) preprocessed, 6 c) segmented, 6 d) Flipped, 6 e) Resized, 6 f) Rotated, 6 g) BPPC extracted, 6 h) DCT feature extracted, 6 i) classified outcome, and 6 j) Grad-CAM.

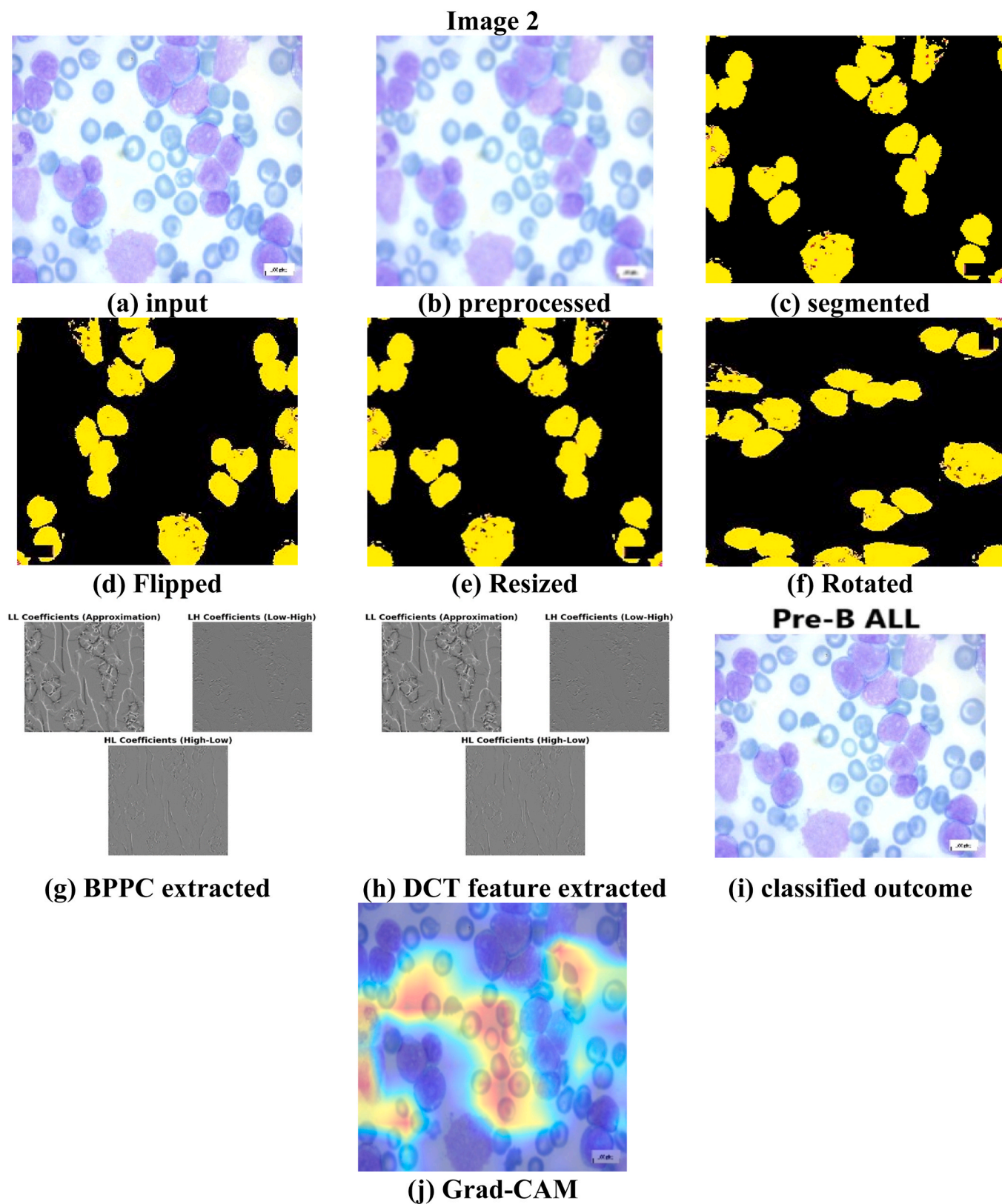


Fig. 7. Experimental results for sample image 2, 7 a) input, 7 b) preprocessed, 7c) segmented, 7 d) Flipped, 7 e) Resized, 7 f) Rotated, 7 g) BPPC extracted, 7 h) DCT feature extracted, 7i) classified outcome, and 7j) Grad-CAM.

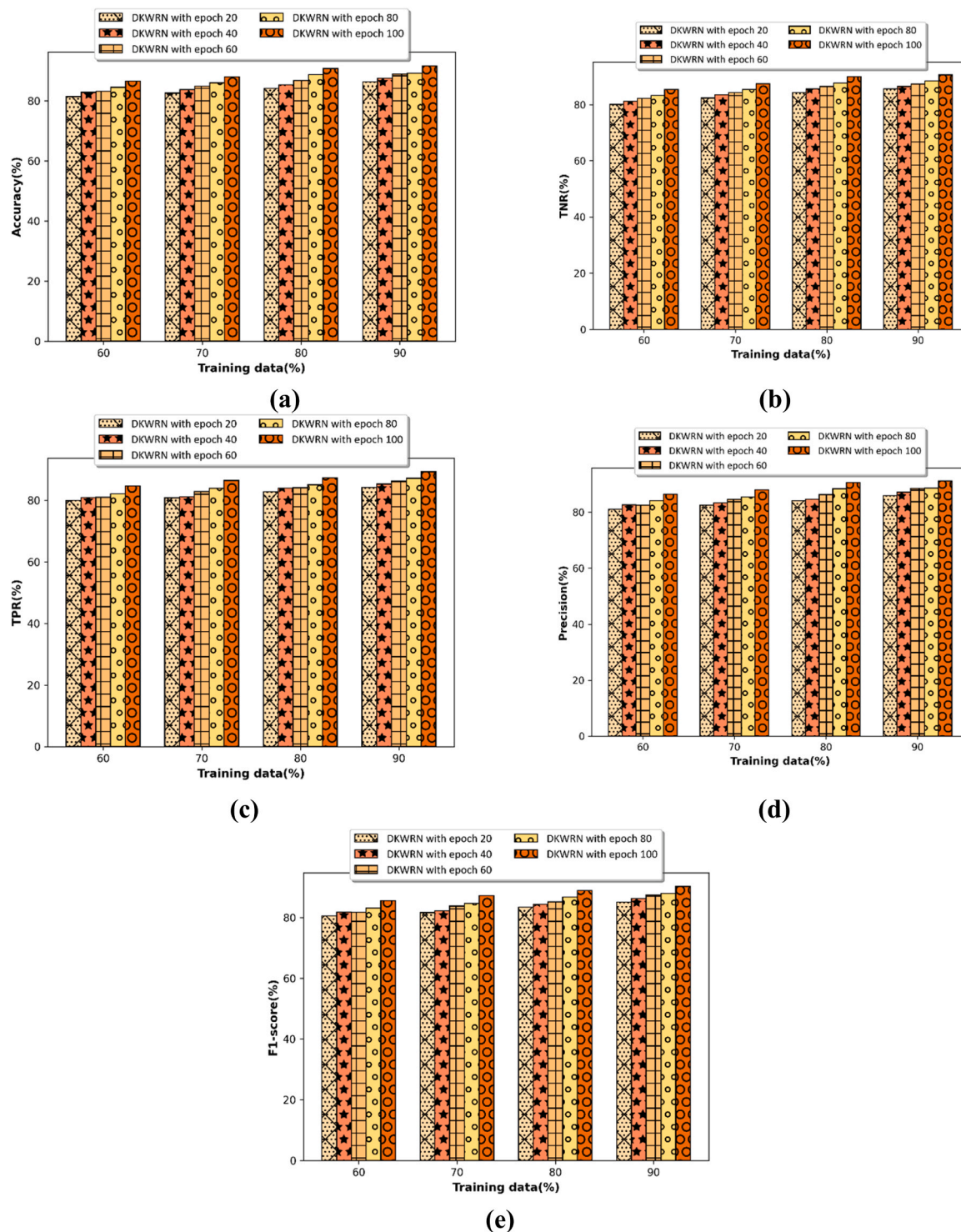


Fig. 8. Performance computation of DKWRN-based Leukemia classification regarding training data, a) Accuracy, b) TNR, c) recall, d) Precision, and e) F1-score.

(Jiang et al., 2020), ResNet-152 (Perveen et al., 2024), Swin Transformer (Prasad and Anbarasi, 2024), DKN (Feng and Yang, 2022), and WRN (Zagoruyko and Komodakis, 2016; Zhang et al., 2021) are used as comparative approaches for validating the effectiveness of DKWRN-based Leukemia classification.

4.7. Comparative evaluation

This subsection outlines a comparative evaluation of the DKWRN-driven Leukemia classification approach, conducted through

variations in training data proportions and adjustments to the K-Fold cross-validation parameters.

4.7.1. Analysis based on training data

The comparative valuation of DKWRN-based Leukemia classification in terms of training data is displayed in Fig. 10. The comparative graph with respect to accuracy is depicted in Fig. 10 a). Considering 60 % training data, the DKWRN attained an accuracy of 85.46 %, in which the BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN attained the value of 80.79 %, 81.53 %, 81.53 %, 81.53 %, 81.53 %, 81.53 %, 81.53 %, and 81.53 %, respectively.

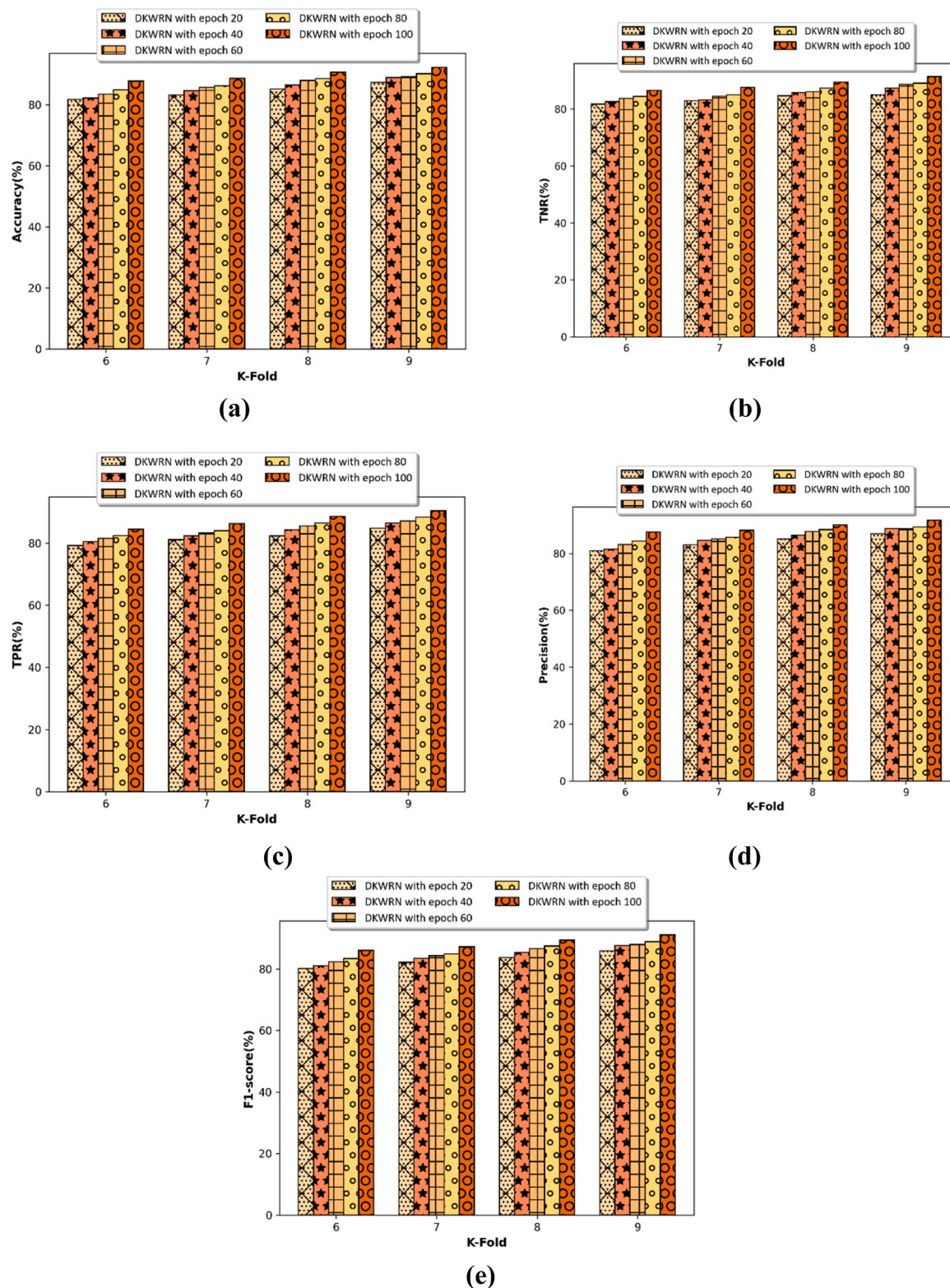


Fig. 9. Performance computation of DKWRN-based Leukemia classification regarding K-Fold, a) Accuracy, b) TNR, c) recall, d) precision, and e) F1-score.

82.34 %, 82.98 %, 83.44 %, 83.78 %, 84.54 %, and 85.09 %, respectively. Therefore, DKWRN obtained 5.46 %, 4.59 %, 3.65 %, 2.36 %, 1.07 %, and 0.43 % of improved performance. Fig. 10 b) shows the analysis in connection with TNR. Using 60 % training data, the TNR of 86.37 % is attained by the DKWRN, whereas the TNR of BO-ALLCNN is 80.40 %, SVM is 81.48 %, ResNet-18 +SVM is 82.39 %, ViT-CNN is 83.05 %, ResNet-152 is 83.76 %, Swin Transformer is 84.32 %, DKN is

85.86 %, and WRN is 86.04 %, respectively. Thus, the performance enhancement of DKWRN compared to the existing approaches is 6.91 %, 5.66 %, 4.60 %, 3.02 %, 0.59 %, and 0.38 %. Fig. 10 c) portrays the evaluation related to recall. While considering 60 % of training data, the DKWRN achieved the recall of 84.51 %, in which the recall of 78.18 %, 79.22 %, 80.74 %, 80.19 %, 81.80 %, 89.67 %, 83.09 %, and 83.74 % are attained using BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN,

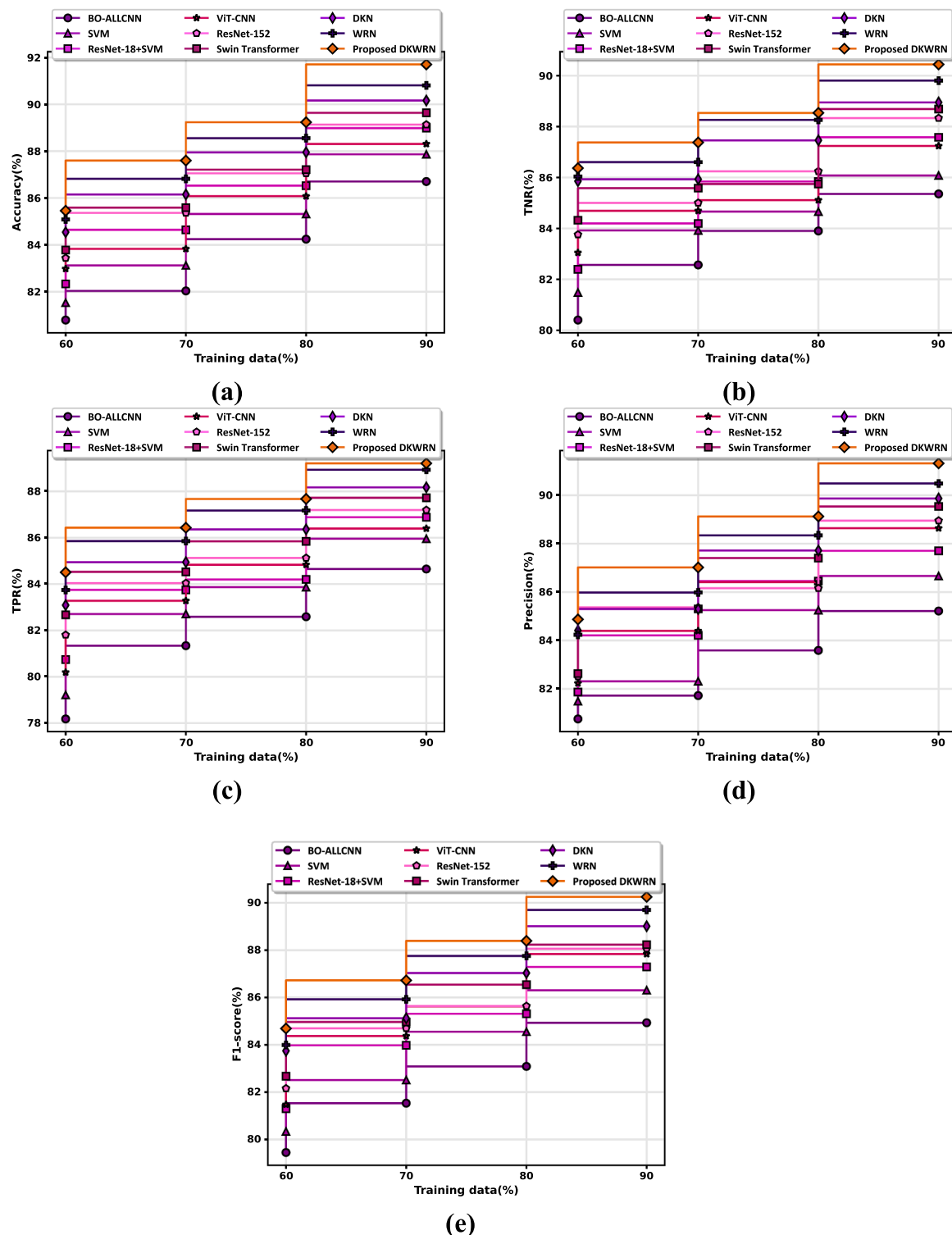


Fig. 10. Comparative analysis of DKWRN-based Leukemia classification regarding training data, a) Accuracy, b) TNR, c) recall, d) precision, and e) F1-score.

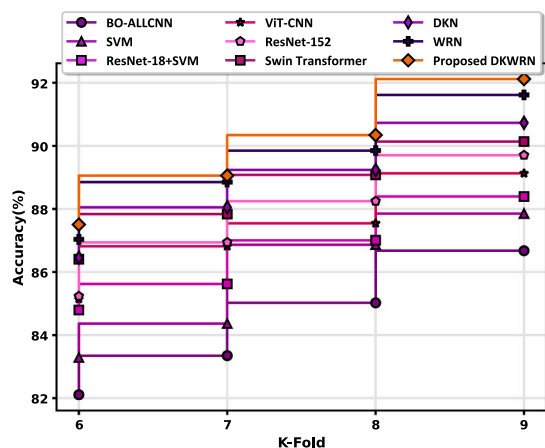
ResNet-152, Swin Transformer, DKN, and WRN. Therefore, the DKWRN achieved 7.49 %, 6.26 %, 4.46 %, 3.20 %, 1.68 %, and 0.90 % of superior performance. The comparative graph with respect to precision is depicted in Fig. 10 d). Considering 70 % training data, the DKWRN attained a precision of 87.02 %, in which the BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN attained the value of 81.72 %, 82.32 %, 84.21 %, 84.40 %, 85.36 %, 85.31 %, 85.30 %, and 85.98 %, respectively. Therefore, DKWRN obtained 6.08 %, 5.40 %, 3.2 %, 1.90 %, 1.97 %, and 1.19 % of improved

performance. Fig. 10 e) shows the analysis in connection with F1-score. Using 80 % training data, the F1-score of 88.39 % is attained by the DKWRN, whereas the score of BO-ALLCNN is 83.09 %, SVM is 84.55 %, ResNet-18 +SVM is 85.31 %, ViT-CNN is 85.62 %, ResNet-152 is 85.64 %, Swin Transformer is 86.54 %, DKN is 87.03 %, and WRN is 87.76 %, respectively. Thus, the performance enhancement of DKWRN compared to the existing approaches is 6.00 %, 4.34 %, 3.48 %, 3.11 %, 1.54 %, and 0.72 %, respectively.

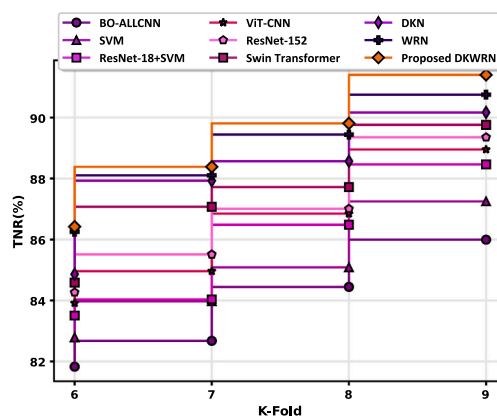
4.7.2. Analysis based on K-Fold

Fig. 11 depicts the comparative estimation of DKWRN-based Leukemia classification regarding different K-Fold. The assessment with respect to accuracy is shown in Fig. 11 a). Under the K-Fold of 6, the accuracy of 87.50 % is attained by the DKWRN, in which the accuracy attained by the BO-ALLCNN is 82.11 %, SVM is 83.29 %, ResNet-18 +SVM is 84.79 %, ViT-CNN is 85.15 %, ResNet-152 is 85.23 %, Swin Transformer is 85.23 %, DKN is 86.45 %, and WRN is 87.03 %, respectively. Thus, the improved performance of 6.17 %, 4.82 %,

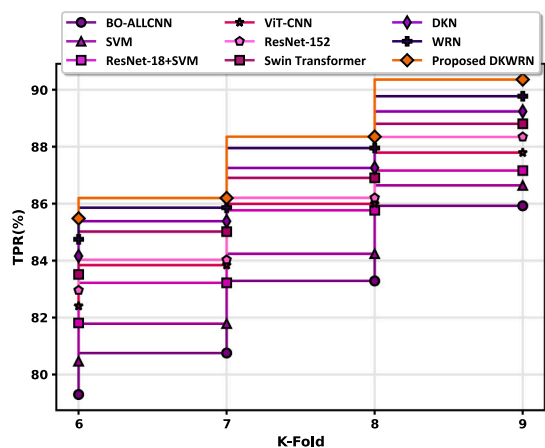
3.10 %, 2.60 %, 1.20 %, and 0.54 % are achieved by the DKWRN while compared to the existing models. Fig. 11 b) deliberates the analysis in connection with TNR. Under the K- Fold = 6, the DKWRN achieved the TNR of 86.42 %, in which the TNR of 81.83 %, 82.79 %, 83.51 %, 83.92 %, 84.27 %, 84.59 %, 84.86 %, and 86.25 % are obtained by the BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN. Thus, the performance improvement of 5.32 %, 4.20 %, 3.37 %, 2.49 %, 1.81 %, and 0.19 % is achieved by DKWRN. The evaluation related to recall is shown in Fig. 11 c). For the



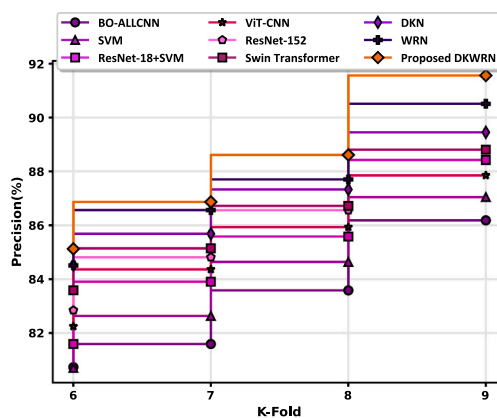
(a)



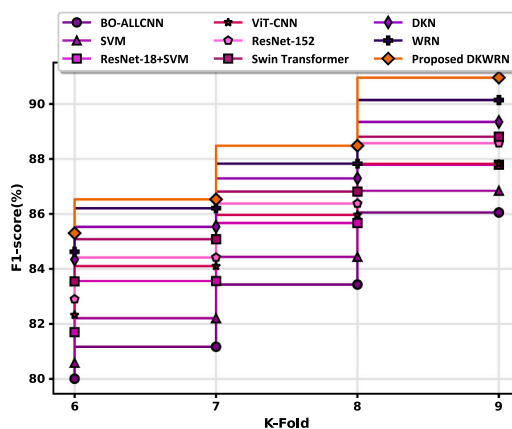
(b)



(c)



(d)



(e)

Fig. 11. Comparative analysis of DKWRN-based Leukemia classification regarding K-Fold, a) Accuracy, b) TNR, c) recall, d) precision, and e) F1-score.

K-Fold of 6, the recall attained by the DKWRN is 85.48 %, and the attained values for BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN are 79.30 %, 80.47 %, 81.81 %, 82.41 %, 82.96 %, 83.51 %, 84.16 %, and 84.75 %, respectively. Hence, the increased performance of 7.23 %, 5.86 %, 4.29 %, 2.95 %, 1.55 %, and 0.86 % is attained by the DKWRN. The assessment with respect to precision is shown in Fig. 11d). Under the K-Fold of 7, the precision of 86.87 % is attained by the DKWRN, in which the precision attained by the BO-ALLCNN is 81.59 %, SVM is 82.63 %, ResNet-18 +SVM is 83.90 %, ViT-CNN is 84.36 %, ResNet-152 is 84.81 %, Swin Transformer is 85.14 %, DKN is 85.68 %, and WRN is 86.56 %, respectively. Thus, the improved performance of 6.07 %, 4.88 %, 3.41 %, 2.37 %, 1.36 %, and 0.35 % are achieved by DKWRN while compared to the existing models. Fig. 11 e) deliberates the analysis in connection with F1-score. Under the K-Fold = 8, the DKWRN achieved the F1-score of 88.48 %, in which the F1-score of 83.43 %, 84.44 %, 85.67 %, 85.96 %, 86.38 %, 86.81 %, 87.29 %, and 87.83 % are obtained by the BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN. Thus, the performance improvement of 5.71 %, 4.57 %, 3.17 %, 2.37 %, 1.34 %, and 0.74 % is achieved by DKWRN.

4.8. Comparative discussion

Table 3 shows the comparative discussion for DKWRN-based Leukemia classification. Here, the results obtained of DKWRN and the existing models such as BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN are displayed. The evaluation measures like accuracy, TNR, and recall compute the performance. Here, the finest values results are achieved in K-Fold-based assessment. The superior accuracy of 92.12 % is achieved by the DKWRN, whereas the achieved value for BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN is 86.67 %, 87.85 %, 88.39 %, 89.13 %, 89.70 %, 90.14 %, 90.73 %, and 91.61 %, respectively. The better TNR attained by DKWRN is 91.36 %, and the TNR of 85.99 %, 87.25 %, 88.46 %, 88.95 %, 89.35 %, 89.76 %, 90.17 %, and 91.40 % is achieved by the existing approaches. The ideal recall of 90.36 % is attained by the DKWRN, whereas the recall of BO-ALLCNN is 85.92 %, SVM is 86.64 %, ResNet-18 +SVM is 87.16 %, ViT-CNN is 87.79 %, ResNet-152 is 88.34 %, Swin Transformer is 88.81 %, DKN is 89.24 %, and WRN is 89.77 %, respectively. Furthermore, the accuracy, TNR, recall, precision, and F1-score of 92.12 %, 90.36 %, 91.40 %, 91.56 % and 90.96 % are obtained by the DKWRN with respect to the K-Fold-Fold values. DKWRN obtained the superior

Table 3
Comparative discussion of DKWRN-based Leukemia classification.

Variations	Metrics/ Methods	BO- ALLCNN	SVM	ResNet- 18 +SVM	ViT-CNN	ResNet- 152	Swin Transformer	DKN	WRN	Proposed DKWRN
Training data= 90 %	Accuracy (%)	86.70	87.87	88.98	88.31	89.13	89.64	90.16	90.81	91.70
	TNR (%)	85.36	86.08	87.58	87.24	88.33	88.69	88.95	89.81	90.44
	recall (%)	84.65	85.96	86.88	86.40	87.19	87.72	88.17	88.93	89.21
	Precision (%)	85.22	86.66	87.70	88.64	88.95	89.54	89.87	90.49	91.32
	F1-score (%)	84.93	86.31	87.29	87.84	88.06	88.23	89.01	89.70	90.25
K-Fold= 9	Accuracy (%)	86.67	87.85	88.39	89.13	89.70	90.14	90.73	91.61	92.12
	TNR (%)	85.99	87.25	88.46	88.95	89.35	89.76	90.17	90.75	91.40
	recall (%)	85.92	86.64	87.16	87.79	88.34	88.81	89.24	89.77	90.36
	Precision (%)	86.18	87.05	88.43	87.85	88.80	88.81	89.45	90.51	91.56
	F1-score (%)	86.05	86.84	87.79	87.82	88.57	88.81	89.35	90.14	90.96
standard deviation										
Variations	Metrics/ Methods	BO- ALLCNN	SVM	ResNet- 18 +SVM	ViT-CNN	ResNet- 152	Swin Transformer	DKN	WRN	Proposed DKWRN
Training data= 90 %	Accuracy (%)	2.44	2.38	2.28	2.25	2.15	2.12	2.10	2.09	2.07
	TNR (%)	1.92	1.82	1.69	1.67	1.60	1.51	1.49	1.48	1.27
	recall (%)	2.44	2.35	2.29	2.18	1.94	1.89	1.87	1.85	1.72
	Precision (%)	2.56	2.40	2.37	2.36	2.29	2.23	2.13	2.11	1.72
	F1-score (%)	2.30	2.23	2.18	2.12	2.12	2.05	2.05	2.02	1.99
K-Fold= 9	Accuracy (%)	1.84	1.72	1.69	1.66	1.65	1.57	1.43	1.39	1.37
	TNR (%)	1.99	1.93	1.92	1.90	1.85	1.83	1.66	1.64	1.61
	recall (%)	2.52	2.36	2.10	2.07	2.05	1.99	1.93	1.92	1.91
	Precision (%)	2.49	2.38	2.36	2.20	2.17	2.10	2.06	1.93	1.85
	F1-score (%)	2.36	2.31	2.28	2.14	2.13	2.05	2.04	1.96	1.89
Confidence Intervals										
Variations	Metrics/ Methods	BO- ALLCNN	SVM	ResNet- 18 +SVM	ViT-CNN	ResNet- 152	Swin Transformer	DKN	WRN	Proposed DKWRN
Training data= 90 %	Accuracy (%)	[80.58, 86.90]	[81.29, 87.99]	[82.19, 89.17]	[82.85, 88.60]	[83.27, 89.40]	[83.67, 89.89]	[84.28, 90.32]	[84.89, 91.08]	[85.31, 91.80]
	TNR (%)	[80.15, 85.57]	[81.28, 86.19]	[82.25, 87.78]	[82.79, 87.45]	[83.48, 88.61]	[84.17, 88.80]	[85.73, 89.17]	[85.85, 90.09]	[86.20, 90.56]
	recall (%)	[77.99, 84.80]	[78.95, 86.22]	[80.48, 87.06]	[80.02, 86.64]	[81.68, 87.38]	[82.54, 87.87]	[82.92, 88.34]	[83.47, 89.12]	[84.21, 89.34]
	Precision (%)	[80.58, 85.35]	[81.29, 86.78]	[81.75, 87.81]	[82.05, 88.93]	[82.31, 89.22]	[82.37, 89.77]	[84.25, 90.06]	[83.97, 90.68]	[84.75, 91.47]
	F1-score (%)	[79.27, 85.22]	[80.21, 86.51]	[81.16, 87.54]	[81.20, 88.07]	[81.98, 88.21]	[82.54, 88.50]	[83.58, 89.27]	[83.86, 89.85]	[84.58, 90.53]
K-Fold= 9	Accuracy (%)	[81.97, 86.88]	[83.14, 88.09]	[84.58, 88.64]	[84.98, 89.31]	[85.02, 89.81]	[86.11, 90.25]	[86.34, 90.94]	[86.92, 91.82]	[87.24, 92.31]
	TNR (%)	[81.58, 86.19]	[82.51, 87.36]	[83.31, 88.68]	[83.78, 89.19]	[84.09, 89.45]	[84.39, 89.87]	[84.58, 90.36]	[86.11, 90.91]	[86.17, 91.55]
	recall (%)	[79.10, 86.14]	[80.34, 86.83]	[81.59, 87.36]	[82.22, 88.03]	[82.66, 88.59]	[83.32, 88.93]	[83.98, 89.45]	[84.52, 89.90]	[85.35, 90.62]
	Precision (%)	[80.45, 86.39]	[80.58, 87.16]	[81.48, 88.55]	[82.06, 87.98]	[82.64, 88.94]	[83.48, 89.03]	[84.24, 89.61]	[84.34, 90.67]	[84.87, 91.79]
	F1-score (%)	[79.77, 86.34]	[80.41, 87.11]	[81.41, 88.07]	[82.16, 87.97]	[82.78, 88.77]	[83.27, 89.04]	[84.06, 89.49]	[84.43, 90.43]	[85.08, 91.26]

outcomes due to its superior processing capability and parameter tuning. Moreover, the proposed DKWRN method demonstrates superior performance over existing methods such as BO-ALLCNN, SVM, ResNet-18 combined with SVM, and ResNet-152 due to its integrated architecture that combines the strengths of WRN and DKN, enabling more effective feature learning and generalization. Unlike traditional models that struggle with the morphological variability of leukemic cells, DKWRN benefits from a robust preprocessing pipeline using adaptive Gaussian filtering, precise segmentation via RefineNet, and comprehensive feature extraction incorporating BPPC, DCT, and statistical features. Furthermore, the class imbalance is mitigated through data augmentation techniques such as flipping, resizing, and rotation, which help enhance minority class representation and improve model generalization. To address noisy slide conditions, adaptive Gaussian filtering is employed during preprocessing to suppress image noise and enhance cell boundaries, ensuring reliable segmentation and feature extraction. These enhancements collectively improve the model's sensitivity to subtle patterns in PBS images. As a result, DKWRN achieves higher classification accuracy and strong performance across other key metrics such as TNR, recall, precision, and F1-score, clearly indicating its effectiveness and reliability compared to the mentioned baseline methods. Additionally, the standard deviation is a fundamental statistical measure that quantifies the amount of variation or dispersion in a set of data values. It reflects how much individual data points deviate from the mean and thus serves as an indicator of consistency within the dataset. A low standard deviation suggests that the values are closely clustered around the mean, indicating stability or predictability, whereas a high standard deviation implies greater spread and variability, often pointing to unpredictability or diverse outcomes. Moreover, a confidence interval is a range of values, derived from sample data, that is likely to contain the true value of an unknown population parameter. It provides an estimate along with a level of certainty.

4.9. Segmentation analysis

The segmentation task is carried out using the RefineNet architecture, and its effectiveness is critically analyzed throughout this portion of the study. The developed RefineNet is compared with models, namely UNet++ (Jan et al., 2025), Scribble2label (Das and Padmanaban, 2025), and U-net (Jiawen and Dahlan, .). Fig. 12 provides insight into the segmentation accuracy metrics obtained using the RefineNet architecture. The segmentation accuracy obtained with a training sample of 90 % by UNet++ is 85.966 %, Scribble2label is 87.324 %, U-net is 89.729 %, and the proposed RefineNet is 92.515 %, respectively. Here,

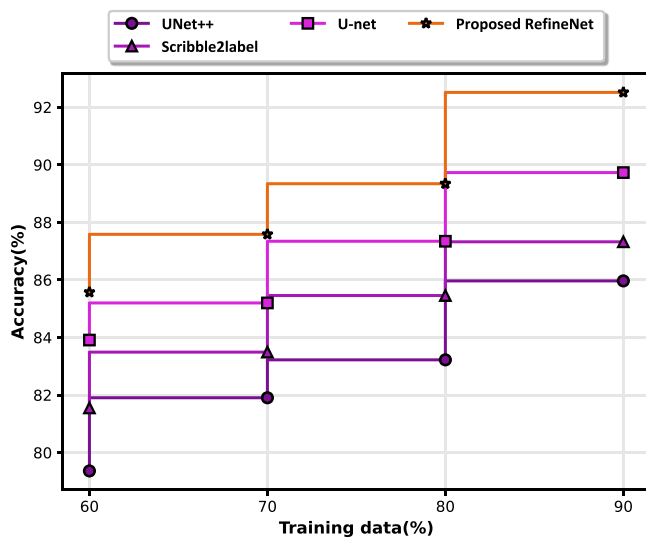


Fig. 12. Analysis of Segmentation.

the segmentation accuracy attained by RefineNet is 5.61 % higher than Scribble2label.

4.10. Feature importance analysis

Fig. 13 represents the proposed method with and without features based on accuracy. When training data is 60 %, the accuracy of DKWRN with BPPC, DKWRN with DCT, DKWRN with statistical features, and Proposed DKWRN attained the accuracy of 81.463 %, 82.967 %, 83.371 %, and 84.600 %. The DKWRN with BPPC, DKWRN with DCT, DKWRN with statistical features, and Proposed DKWRN attained the accuracy values of 86.311 %, 88.527 %, 90.402 %, and 91.675 %, respectively, when training data is 90 %. This progression highlights the positive correlation between training data volume and model performance, with the proposed DKWRN consistently outperforming other variants across both data levels due to its enhanced feature integration and learning capacity.

4.11. Statistical analysis

4.11.1. ANOVA test

Statistical testing is crucial for evaluating experimental results and determining whether the data support a hypothesis for broader applications. A key factor is the P-value, which measures the likelihood of obtaining results if the null hypothesis is true. A P-value below 0.05 typically indicates significance, justifying the rejection of the null hypothesis. In this study, ANOVA is used to assess whether differences in performance between methods are statistically significant. As presented in Table 4, the precision, accuracy, TNR, TPR, and F1-score based ANOVA results yielded P-values below 0.05 when assessed using accuracy as a metric, confirming that the enhancements introduced by the proposed approach are statistically significant.

4.11.2. t-test

To determine whether the proposed approach yields statistically meaningful improvements over baseline models, paired *t*-tests are employed. This evaluation is based on five key performance indicators: Accuracy, TNR, TPR, Precision, and F1-score derived from 90 % of the training dataset. For each metric, the method's performance across individual folds is systematically compared to that of the reference models. A threshold of $p < 0.05$ is used to denote statistical significance. The outcomes of these comparative tests, which highlight the robustness of the proposed technique, are summarized in Table 5.

The evaluation reveals that the newly introduced approach consistently surpasses all benchmark models, including BO-ALLCNN, SVM,

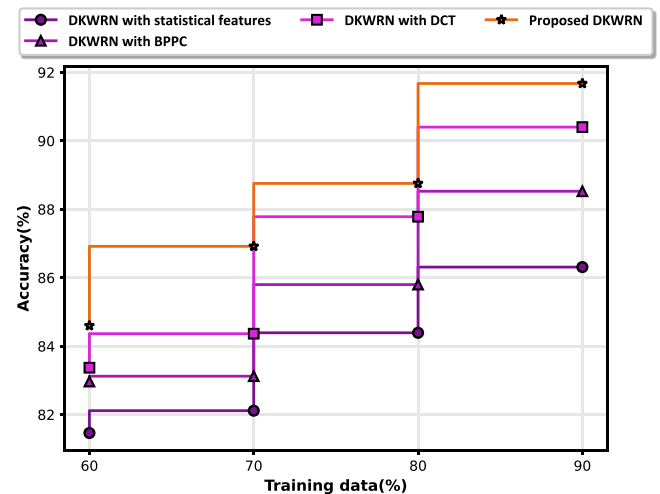


Fig. 13. Feature Analysis.

Table 4
ANOVA test.

Metrics	Source	Sum of squares	Degrees of freedom	F	P-value
Precision	C (treatments)	87.3498	3	417.2941	5.27e-23
	Residual	2.2384	32	-	-
Accuracy	C (treatments)	80.0736	3	294.5117	1.23e-21
	Residual	2.6054	32	-	-
TNR	C (treatments)	86.7112	3	401.9236	9.42e-23
	Residual	2.3038	32	-	-
TPR	C (treatments)	90.1456	3	437.2384	2.86e-23
	Residual	2.2065	32	-	-
F1-Score	C (treatments)	87.8029	3	418.9334	4.87e-23
	Residual	2.2381	32	-	-

ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN, across five critical performance indicators: Accuracy, TNR, TPR, Precision, and F1-score. Each comparative analysis yields statistically significant outcomes ($p < 0.05$), underscoring the reliability of the results. In terms of accuracy, the proposed method achieves notably higher scores than BO-ALLCNN ($t = 2.73$, $p = 0.03$), SVM ($t = 2.33$, $p = 0.03$), ResNet-18 +SVM ($t = 2.30$, $p = 0.03$), ViT-CNN ($t = 2.16$, $p = 0.03$), ResNet-152 ($t = 1.85$, $p = 0.01$), Swin Transformer ($t = 1.77$, $p = 0.01$), DKN ($t = 1.57$, $p = 0.01$), and WRN ($t = 1.30$, $p = 0.01$). Comparable improvements are observed across the remaining metrics such as TNR, TPR, Precision, and F1-score, reinforcing the method's strength in delivering reliable and enhanced classification outcomes over conventional techniques.

4.12. Confusion matrix

Fig. 14 shows the confusion matrix for the dataset. A confusion matrix also referred to as an error matrix in the realm of statistical classification serves as a foundational tool for assessing the performance of classification algorithms. In unsupervised learning, it's sometimes known as a matching matrix, where true labels are not available and the matrix helps evaluate how predicted clusters align with ground truth categories post hoc. Here, all matrices are calculated using confusion matrix.

4.13. Training time and Inference Time

The training time of the methods is depicted in Table 6. Here, the table compares various methods for Leukemia classification in terms of their training time. The methods listed include BO-ALLCNN, SVM, ResNet-18 +SVM, ResNet-152, DKN, WRN, and Proposed DKWRN. The training times (in minutes) for each method are reported, with the proposed DKWRN achieving the shortest training time of 3.174 min.,

Table 5
t-test.

Proposed method	Comparative methods	Dataset 1									
		T-statistic					P-value				
		ACC	TNR	TPR	PRE	F1	ACC	TNR	TPR	PRE	F1
Proposed DKWRN	BO-ALLCNN	2.73	2.72	2.92	2.31	2.30	0.03	0.04	0.03	0.03	0.04
	SVM	2.33	2.51	2.76	2.09	2.09	0.03	0.03	0.03	0.03	0.04
	ResNet-18 +SVM	2.30	2.12	2.72	2.00	2.05	0.03	0.03	0.03	0.03	0.04
	ViT-CNN	2.16	2.06	2.24	1.84	2.04	0.03	0.03	0.03	0.02	0.04
	ResNet-152	1.85	1.56	2.01	1.54	1.63	0.01	0.02	0.03	0.01	0.03
	Swin Transformer	1.77	1.53	2.00	1.53	1.62	0.01	0.02	0.03	0.01	0.03
	DKN	1.57	1.45	1.81	1.44	1.45	0.01	0.02	0.02	0.01	0.02
	WRN	1.30	1.37	1.73	1.43	1.17	0.01	0.02	0.02	0.01	0.02

per full training cycle, demonstrating its superior efficiency compared to the other methods. Moreover, the inference time results in the table highlighting the efficiency of various methods in processing input data, measured in milliseconds. The Proposed DKWRN model achieves the lowest inference time of 0.135 ms, showcasing superior efficiency likely due to the integration of DKN's lightweight design with WRN's fast and wide structure, making it highly suitable for real-time applications.

4.14. Cross-dataset validation

Fig. 15 shows the Cross-dataset Validation using the ALL-image datasets for training and Leukemia-Classification Computer Vision Dataset for testing. When training data is 60 %, the accuracy of the proposed method is 83.988 %, whereas, the traditional methods such as BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN attained the accuracy of 78.939 %, 79.932 %, 80.479 %, 81.312 %, 81.907 %, 82.090 %, 82.517 %, and 83.171 %. Similarly, when the training data is 90 %, BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, WRN, and the proposed DKWRN method attained the accuracy of 84.030 %, 84.725 %, 85.215 %, 86.449 %, 87.083 %, 87.986 %, 88.623 %, 89.181 %, and 89.889 %. This progression underscores the effectiveness of deeper architectures and hybrid strategies, with DKWRN demonstrating superior generalization and feature learning capabilities.

4.15. Validation analysis

Training and validation accuracy of the proposed model over 100 epochs is presented in Fig. 16. A Validation Curve is a graphical tool

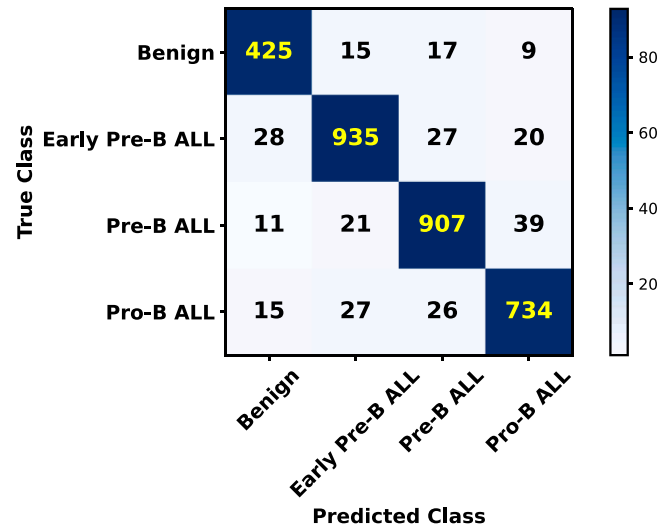


Fig. 14. confusion matrix.

Table 6
Training Time.

Methods	BO-ALLCNN	SVM	ResNet-18 +SVM	ViT-CNN	ResNet-152	Swin Transformer	DKN	WRN	Proposed DKWRN
Training time (min)	8.323	7.871	7.147	6.965	6.612	6.135	5.574	4.872	3.174
Inference Time (ms)	0.617	0.539	0.449	0.378	0.329	0.295	0.247	0.189	0.135

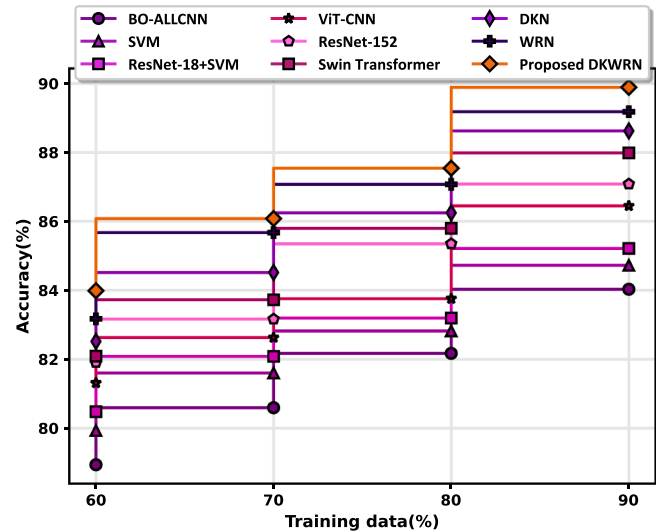


Fig. 15. Cross-dataset validation.

used in machine learning to evaluate how the performance of a model varies with changes in a specific hyperparameter. The graph illustrates model performance over an iteration range of 0–100 epochs, revealing a clear upward trend in accuracy as training progresses. Specifically, as the number of epochs increases from 0 to 100, the training accuracy improves significantly from 50 % to 92.2 %, while the validation accuracy rises from 50 % to 90.5 %. This consistent growth indicates effective learning and generalization of the model over time.

Training and validation loss of the proposed model over 100 epochs is given in Fig. 17. From the graph, as the epoch increases, the loss value decreases. As training progresses from 0 to 100 epochs, the model's training loss drops sharply from 0.950 to 0, indicating strong learning, while validation loss decreases from 0.990 to 0.109, reflecting improved generalization.

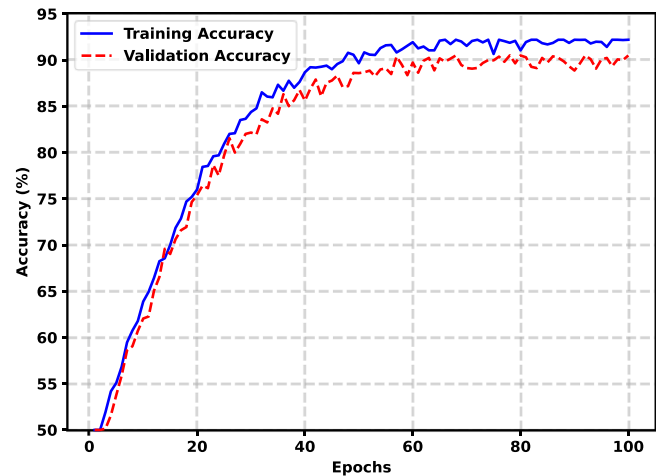


Fig. 16. Training and validation accuracy of the proposed model over 100 epochs.

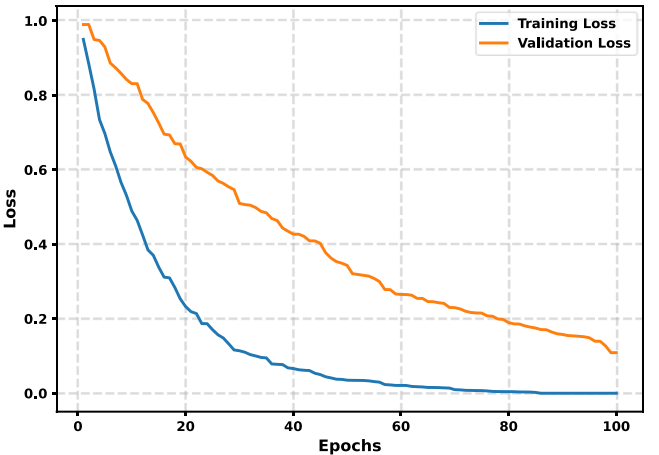


Fig. 17. Training and validation loss of the proposed model over 100 epochs.

4.16. Scalability analysis

Scalability analysis is the process of evaluating how well a system, model, or process handles increasing workloads, data volumes, or user demands without compromising performance, reliability, or efficiency. It helps determine whether a solution grows and adapts as needs expand. Fig. 18 shows the analysis of the scalability. When the image size is 32*32, the accuracy of the proposed DKWRN is 85.319 %, whereas the traditional methods, such as BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN, attained an accuracy of 80.454 %, 81.389 %, 82.089 %, 83.387 %, 83.608 %, 84.059 %, 84.549 %, and 84.835 %. This improvement highlights DKWRN's superior ability to extract meaningful features even from low-resolution inputs. Similarly, the runtime of existing methods like BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, WRN, and the proposed DKWRN method is 87.168 %, 88.073 %, 88.359 %, 89.054 %, 89.416 %, 90.010 %, 90.609 %, 91.428 %, and 91.869 %, when the image size is 256*256. These results

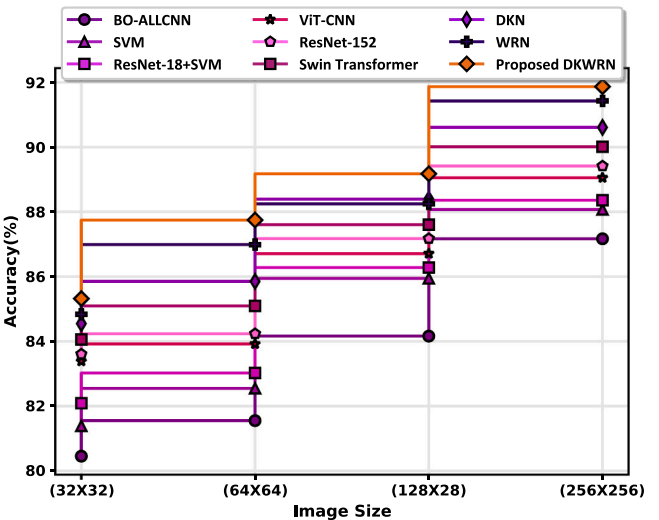


Fig. 18. Scalability Analysis.

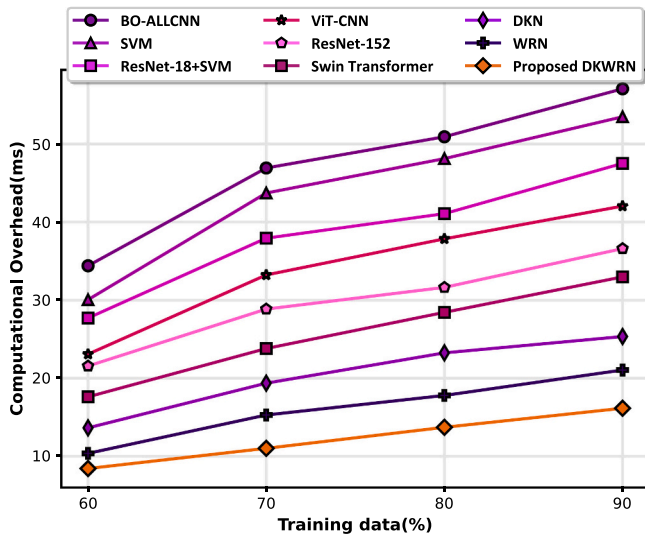


Fig. 19. Computational Overhead.

justify the robustness and scalability of DKWRN across varying input dimensions, balancing both accuracy and computational efficiency.

4.17. Computational overhead

Fig. 19 illustrates the analysis of computational overhead. In this graph, when training data is 60 %, the proposed DKWRN method attained the computational overhead of 8.447 ms, whereas the traditional methods, such as BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, and WRN, attained the computational overhead of 34.407 ms, 30.063 ms, 27.714 ms, 23.088 ms, 21.569 ms, 17.622 ms, 13.652 ms, and 10.385 ms. Similarly, when training data is 90 %, the computational overhead of the BO-ALLCNN is 57.067 ms, SVM is 53.484 ms, ResNet-18 +SVM is 47.518 ms, ViT-CNN is 42.047 ms, ResNet-152 is 36.604 ms, Swin Transformer is 32.983 ms, DKN is 25.348 ms, WRN is 21.056 ms, and the proposed DKWRN is 16.164 ms. These results highlight DKWRN's scalability and consistent performance gains, making it a highly efficient choice for computationally demanding tasks.

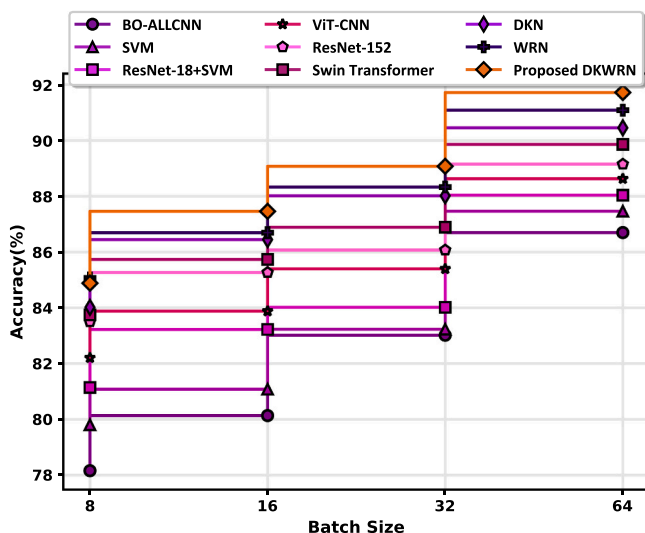


Fig. 20. Comparison of baseline models by varying the hyperparameters.

4.18. Analysis by varying the Hyperparameter

Fig. 20 presents a comparative analysis of model performance across varying batch sizes, highlighting the impact of hyperparameter tuning on classification accuracy. At a batch size of 8, the proposed DKWRN model achieved an accuracy of 84.882 %, whereas the accuracy of traditional methods, such as BO-ALLCNN, is 78.160 %, SVM is 79.807 %, ResNet-18 +SVM is 81.143 %, ViT-CNN is 82.207 %, ResNet-152 is 83.502 %, Swin Transformer is 83.747 %, DKN is 84.016 %, and WRN is 85.067 %. This suggests that DKWRN is highly competitive even under constrained batch conditions. Likewise, when batch size is 64, the accuracy of BO-ALLCNN, SVM, ResNet-18 +SVM, ViT-CNN, ResNet-152, Swin Transformer, DKN, WRN, and the proposed DKWRN method is 86.696 %, 87.461 %, 88.036 %, 88.627 %, 89.149 %, 89.859 %, 90.451 %, 91.083 %, and 91.718 %. These results justify DKWRN's design enhancements, showing that it not only maintains strong performance at lower batch sizes but also scales effectively with larger batches, making it a reliable choice for high-throughput training scenarios.

5. Conclusion

Leukemia is considered as one of the most hazardous forms of cancer, which affects the person in all age groups, including adults, and children. Leukemia is the abnormal growth of immature lymphocytes. To detect leukemia, image processing approaches are considered to estimate microscopic PBS images. Moreover, the detection and classification of Leukemia is necessary to provide proper treatment as per the disease classes. However, it is complicated due to the varying size, and texture of blood cells even in similar types of Leukemia. Hence, the DKWRN-based Leukemia classification is proposed in this paper. The adaptive Gaussian filter eliminates the noise in the preprocessing stage, and the RefineNet segments the Blast cell from the image. Moreover, augmentation enhances the size of the image, and the significant features are extracted in the feature extraction. Moreover, Leukemia is classified into early Pre-B, Pre-B, Pro-B, and Hematogones by the utilization of DKWRN. Moreover, the DKWRN-based Leukemia classification achieved superior accuracy, TNR, recall, precision, and F1-score of 92.12 %, 91.40 %, 90.36 %, 91.56 %, and 90.96 %, respectively. Moreover, the performance improvement of proposed DKWRN over the traditional methods, such as BO-ALLCNN, SVM, ResNet-18 +SVM, ResNet-152, DKN, and WRN, are 5.91 %, 4.63 %, 4.04 %, 2.27 %, 1.51 %, and 0.55 % respectively. The biomedical implications of the proposed DKWRN model are highly significant, as it provides an efficient and reliable computer-assisted diagnostic framework for the early detection and classification of ALL subtypes from peripheral blood smear images. Accurate identification of early Pre-B, Pre-B, Pro-B, and Hematogones stages is clinically important, as it supports hematologists in differentiating between malignant and normal precursor cells, thereby aiding in precise diagnosis and appropriate treatment planning. By automating the diagnostic process, the model reduces manual effort, minimizes human error, and accelerates clinical decision-making, particularly in laboratories with limited expert availability. However, the effectiveness of the model is limited in low-resource environments due to its dependence on well-annotated, high-quality blood smear images, which are often unavailable or inconsistently produced in such settings. The future work will focus on developing a hybrid optimization strategy aimed at minimizing processing time by jointly optimizing the kernelization process in the DKN and the depth-width trade-off parameters in the WRN. Moreover, collaboration with hematologists to assess model interpretability, workflow compatibility, and diagnostic reliability will be crucial for its acceptance in medical practice.

The code of this approach is present in <https://github.com/MylapaliRamesh/DKWRN.git>.

CRediT authorship contribution statement

Mylapalli Ramesh: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing - original draft, Writing - review & editing, Project administration. **Naga Mallikharjunarao Billa:** Conceptualization, Methodology, Software. **Dr.Venkata Kishore Kumar Rejeti:** Validation, Formal analysis, Investigation. **Cherukuri Phanindra:** Resources, Data Curation, Writing - original draft, Writing - review & editing, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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